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Comminution mechanisms, particle shape evolution and collision energy partitioning in tumbling mills

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ABSTRACT

A computational model for use with DEM (Discrete Element Method) is proposed for the five mechanisms applying to particle comminution in AG, SAG and coarse feed ball mills. Chipping and rounding are mechanisms that lead to preferential mass loss from corners and edges of particles and which produce shape change for the particles as well as size reduction. These are controlled by the energy dissipation at a contact and the location of the contact on the surfaces of the particles. These mechanisms lead to rounding or conditioning of angular feed particles. Single impact body breakage is a very weak contributor to overall size reduction with most body breakage occurring via damage accumulation over many contacts. This incremental damage mechanism is inherently less inefficient but leads to size reduction from the many weak collisions experienced by particles within the mill. Finally, size reduction from abrasion is well represented by the shear energy absorption of the particles. Particle size and shape evolution due to chipping, rounding and attrition is demonstrated using a well characterised pilot mill for which detailed particle mass data is available. The relative contributions of the five mechanisms and a quantification of the wasted energy for both AG and SAG charges and for new and substantially comminuted material are reported. Changes in the energy spectra with the decreasing particle sizes over time are described with increases in the fraction of collisions above the elastic energy threshold leading to faster damage accumulation and reduced energy wastage. Finally, it is shown that the attribution of dissipated energy between particles in a collision is material dependent with significantly more energy absorbed by rock particles than balls. This needs to be accounted for in DEM models, particularly when attempting to explicitly predict particle size reduction.

1. Introduction

The Discrete Element Method (DEM) is a computational scheme for predicting the flow of particulates. Since milling is predominately a process involving collisions between particles and between particles and the milling machine, it has been used extensively to investigate different aspects of mill performance. The earliest use for modelling in SAG mills was by Rajamani and Mishra (1996) who used two dimensional DEM to help design lifters by predicting media overthrow. The most common use has been in understanding the charge distribution, material flow patterns, energy utilisation and power draw (Mishra and Rajamani, 1992, 1994; Cleary, 1998, 2001a, 2001b, 2001c; Herbst and Nordell, 2001; Cleary, 2004, 2009; Morrison and Cleary, 2004, 2008; Cleary et al., 2008 and many authors since then including for example the comminution special issue edited by Cleary and Morrison, 2008). A recent review article by Weerasekara et al. (2013) describes the current status of DEM modelling and the challenges related to its use in comminution.

One of the most important aspects of mill performance that has been least studied using DEM is that of particle breakage. Although this is the purpose of milling, the use of DEM for the prediction of size reduction is challenging because several different modes of breakage are active and because the particle change (size and shape) needs to be explicitly included in the DEM model. Commonly DEM simulation is used to provide average process data from which selection functions can be determined for use in traditional population balance models (Tuzcu and Rajamani, 2011; Tavares and de Carvalho, 2009). The accuracy of such models is limited by the ability to adequately characterise the collisional environment and the limited physics captured by the population balance models. To be able to accurately predict, from first principles, breakage, product size distribution and resident particle size

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distribution, particle breakage needs to be included directly in the DEM model. The breakage model and the mechanisms that they implement also need to be sufficiently correct to make accurate predictions. This avoids the averaging over the different flow conditions within different parts of the mill that is inherent in approaches that seek to predict particle populations using averaged selection functions. An approach for direct inclusion of breakage in DEM was first proposed by Cleary (2001b). This used a replacement strategy where compressive or impact breakage fracture of a parent particle into finer progeny was represented directly in the DEM model. This model was limited by the use of spherical parent and progeny and the limited ability to include characterised breakage properties. Recently, a more sophisticated variant that can represent breakage of non-round particles into non-round progeny was proposed by Delaney et al. (2010a). This was specifically developed to be able to model incremental breakage due to damage accumulation of particles over many collisions and uses breakage characterisation data from tests such as the JKDWT (JK Drop Weight Test) or JKRBT (JK Rotary Breakage Tester).

At SAG 2006, we reported (Morrison et al., 2006a) the simulated and measured outcomes of treating a well characterised ore in a 1.2 m diameter mill. This well instrumented, pilot scale mill at the University of KwaZulu-Natal had been combined with some new approaches to ore testing to allow different modes of breakage to be tested. It showed that autogenous mill loads of various sizes and shapes could be reasonably predicted by abrasive mass loss proportional to the estimated frictional energy experienced by each particle. However, this approach was inadequate for SAG operation where incremental damage produces more non-trivial body breakage and quite different progeny size distributions. Another drawback of this initial investigation was that the particles were spherical in the DEM models of both the abrasion mill and the pilot mill.

At SAG 2011, we extended this work (Morrison et al., 2011; Delaney et al., 2013) by exploring the accuracy of the breakage prediction when using just an incremental damage mechanism. This used the elastic energy of contacts to estimate the incremental damage that controls the probability of a particle breaking and the progeny produced. Defining E_0 to be the elastic threshold energy at which damage starts to occur, it was found that a range from 3.6 to 5.4 J/kg was needed to produce appropriate amounts of breakage. The model was able to reproduce with good accuracy both AG and SAG product size distributions. In using only the incremental damage, the E_0 values were found to be around an order of magnitude lower that typically measured (Morrison et al., 2007; Whyte, 2005). Much of the size reduction was found to occur from very weak collisions that removed very small mass increments from the particles rather than by substantive body breakage. Essentially, the incremental damage model was also directly predicting attrition of the particles, although little correlation was found between the predicted and measured fine progeny.

In this paper, we identify five key breakage mechanisms and provide details of the computational method for implementing them. We then focus on the surface damage mechanisms by explicitly including the chipping and rounding of non-spherical parent particles in addition to the abrasion mechanism that has been previously reported by Morrison et al. (2006a). This allows us to explicitly predict the size and shape evolution of the particles in the mill resulting from surface damage accumulated over time from their collisions. The surface mass loss rate as a function of energy input is calibrated from one identified rock in the first experiment and this is then used in simulating the evolution of all the particles in three test cases (with differing test conditions). The model predictions using only surface damage form an upper limit on the size of each particle at each time. The nature of the change of shape of the particles during milling is explored in detail as is the variation of these rates with time as the particles become rounder. The decrease in particle size over time will be shown to change the collisional environment in the mill. It will be further shown that as the particles shrink, their specific collision energies increase leading to more collisions being above on the elastic threshold E_0 which will lead to increasing efficiency of the incremental breakage mechanism. Finally, the attribution of collision energy absorption between colliding entities will also be shown to be important and bounds on the splits required to match experiment will be established.

2. Breakage mechanisms in mills

The first few minutes of operation with a fresh charge causes rapid rounding of initially angular particles and a relatively coarse product. The production rate of progeny then decreases to a more or less steady state production rate of much finer progeny. The rate of particle wear can be modelled based on inter-particle friction and shear energy absorption. Under suitable conditions, the number of impacts which exceed the elastic threshold E_0 allows the ore particles to begin to accumulate incremental damage.

Previously (Morrison et al., 2011), postulated five mechanisms occurring in AG, SAG mills and coarser feed ball mills:

- Body breakage (single impact breakage through the particle – the traditionally conceived mechanism occurring in tumbling mills but in reality infrequent in a large SAG mill and completely absent in the pilot mill used here).
- Incremental damage (body breakage due to accumulated damage or fatigue from many weak collisions).
- Attrition or abrasion (mass loss at the surface of rounded rocks as other particles slide over them or they slide over the liner).
- Rounding (preferential and higher abrasive mass loss from sliding at the corners and edges of blocky particles).
- Chipping (loss of corners, edges and larger asperities from small scale body breakage for irregular shaped or non-round particles).

All five mechanisms have been implemented in the DEM code described in Cleary (2004, 2009). For the results reported in this paper, only the last three surface erosion mechanisms are active. The final stage of this work where all the mechanisms are used together to provide a complete representation of the breakage environment experienced by the rock particles will be reported in a following paper. This should, in principle, allow direct prediction of the complete rock size distribution over time.

3. Implementation of size reduction in DEM

3.1. Determination of energy absorbed by a particle from DEM

The contact force is best represented in a collision frame with a normal direction and a tangent plane. This is defined locally at the contact point between the two colliding surfaces. In this frame, using a linear spring–dashpot contact model, the normal force F_n consists of a linear spring to provide the repulsive force and a dashpot to dissipate a specified proportion of the relative kinetic energy:

$$F_n = -k_n \Delta x + C_n v_n. \tag{1}$$

The magnitude of the overlaps Δx between particles is determined by the stiffness k_n of the spring in the normal direction (in N/m). The damping term is proportional to the normal component of the contact velocity v_n. The normal damping coefficient C_n is chosen to give the required coefficient of restitution ε (defined as the ratio of the post-collisional to pre-collisional normal component of the relative velocity), and is given in Schwager and Pöschel (2007) in a more DEM friendly implementation form in Thornton et al. (2013). In many DEM application simulations including in the ones presented here, the stiffness is used as a numerical rather than a physical parameter and is chosen to be larger than the real material stiffness so as to maximise the timestep and therefore minimise the simulation time. This is generally possible as long as the numerical contact deformations remain small and the contact force networks are not altered by the moderate softening of the springs. Typically, average overlaps of 0.1–0.5% are adequate to ensure a balance between computing speed and realistic predictions.

The tangential force is given by:

$$F_t = \min\left\{\mu F_n, \sum k_t v_t \Delta t + C_t v_t\right\}, \quad (2)$$

where the vector force F_t and tangent velocity $\boldsymbol{v}_t$ are defined in the plane tangent to the surface at the contact point. k_t is the tangential spring stiffness (N/m), C_t is the tangential dashpot coefficient and Δt is the integration timestep. The summation term represents an incremental spring that stores energy from the relative tangential motion and models the elastic tangential deformation of the contacting surfaces, while the dashpot dissipates energy from the tangential motion and models the tangential plastic deformation of the contact. The total tangential force F_t is limited by the Coulomb frictional limit μF_n, at which point the surface contact shears and the particles begin to slide over each other. Here μ is the dynamic friction coefficient of the contacting surfaces. The tangential stiffness used is half the normal stiffness.

In terms of choice of contact model Thornton et al. (2013) demonstrated that all inelastic contact models behaved broadly as two classes depending on whether the inelasticity was plastic or viscous in nature. Both classes produce similar types of single particle rebound kinetics at higher coefficients of restitution but increasing different behaviour at moderate to low values. It is an open question as to which class of contact model best represents collisions of different materials, but it appears less likely that elastoplastic models will be appropriate for representing rock collisions which are generally not plastic in nature. For brittle rocks, much of the dissipation arises from small scale fracture at contact points and from spalling. More fundamental investigations of the contact forces for elastic and elastoplastic particles can be performed using the FEM (Wu et al., 2003), but these methods are not suitable for direct body breakage predictions where large deformations, fragmentation and fragment collisions all occur. For these problems the SPH methods formulated for brittle fracture (see Das and Cleary, 2010 for such examples) are better suited for determining how brittle particles break.

Energy dissipation during a collision can be calculated as the integral of the dot product of the dissipative force components (in the collision frame) with the relative contact velocity vector (also in the collision frame). For the linear spring–dashpot model used in this work, the normal component of the dissipative force is the normal dashpot in Eq. (1). The tangential contribution is more complex as the calculation depends on whether the contact is sliding or not. If the contact is not sliding then the dissipative component is the tangential dashpot in Eq. (2) while for the sliding case it is the entire tangential force F_t. The normal component of the energy dissipation is the integral of normal dashpot force with the normal velocity over the collision and the tangential component of energy dissipation is the integral of the tangential dissipative force with the tangential velocity. The total energy dissipation in a collision is then the sum of these two components. It is common practice to consider that body breakage is controlled by the normal energy dissipation while the abrasion is controlled by the tangential energy dissipation. This is based on the normal energy being directed into the body of the particles while the tangential energy is directed in the shear direction and will affect primarily the asperities that contribute to the generation of friction.

The best test of the correctness of the energy dissipation calculations is to integrate these over all collisions to estimate the total energy dissipation in the granular system over the simulated time. For mill simulations, the power draw is accurately given by the product of the torque applied to the mill by all the particle contacts by the mill rotational speed. The total energy supplied by the mill is just the instantaneous power draw integrated over time. For long duration mill simulations that have reached steady state operation, the energy dissipated and the energy supplied by the mill should be equal. In practice, balances within a few percent indicate an accurate DEM simulation. When using the energy dissipated in collisions to predict energy spectra which are used as proxies for understanding breakage or for explicit breakage prediction (as we will do in this paper) it is critically important that these energies are accurate.

The energy dissipated in a collision is the energy that is dissipated between the colliding surfaces of the two bodies. The final step in this energy calculation process is to attribute the energy to each of the bodies involved in the collision. In the absence of better information to the contrary general practice in DEM appears to be to attribute the energy equally to each body. For collisions of similar materials (such as rock particles with rock particles) this is not necessarily an unreasonable assumption. For rock particles impacting the liner or with steel balls, this assumption has no particular foundation and will be a question that we start to address in this paper. This question of energy dissipation between bodies is critical for DEM modelling that includes breakage since it controls the evolution of the particles. When breakage is not explicitly included then this split affects only the energy spectra and any conclusions that are drawn from these.

3.2. Attributing energy between mechanisms and particle shape attributes

The super-quadric particle shape representation is described in Cleary (2004, 2009) which contain examples of use in industrial applications and an explanation of the importance of including shape within DEM simulation). A super-quadric is defined in a canonical frame with Cartesian coordinates (x, y, z) as:

$$\left(\frac{x}{a}\right)^m + \left(\frac{y}{b}\right)^m + \left(\frac{z}{c}\right)^m = 1 \quad (3)$$

The power m is called the blockiness and controls how sharp the corners and edges of the particle are. The semi-major axes in the principle directions are a, b and c and their ratios determine the aspect ratios of the particles with $A_{xy} = b/a$ and $A_{xz} = c/a$. This shape description is particularly well suited to applications where the shape of the particles evolves over time. Since the shape representation is continuous, the shape parameters (m, a, b and c) can be varied incrementally and continuously with the degree of surface erosion closely replicating the real physical behaviour without any artefacts arising from the nature of the numerical shape representation. In contrast, alternative approaches using either polyhedral shapes which have sharp or simply rounded edges or bonded spheres where the curvature of edges and corners is limited by the radius of the sub-particles (which cannot be easily be varied), the super-quadric shape easily supports continuous shape change. Some examples of equi-axed (that is ones with aspect ratios of unity) super-quadric particles are shown in Fig. 1. For $m = 2$ the particles are spherical (as used for the steel balls in SAG case 3). For $m = 3$ the particles are mid-way between spherical and blocky with quite rounded edges and corners. In contrast, for $m = 5$ the

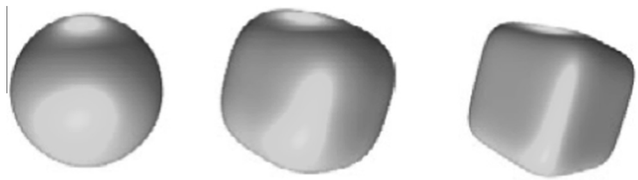

Fig. 1. Super-quadric particle shapes, (a) spherical ($m = 2$), (b) mid-way from spherical to blocky ($m = 3$), and (c) moderately rounded cubical particle ($m = 5$).

particles are reasonably blocky. For higher values of m the edges become increasingly sharp with $m = 10$ giving very cubical particles.

Fig. 2 shows a schematic of the chipping and body breakage energy calculation using information from the collision such as the contact normal $\boldsymbol{N}$ and the moment vector $\boldsymbol{D}$ from the contact point to the particle centre. These are used to apportion the normal energy dissipation between chipping and body breakage (by either high energy single impact collisions such as in a vertical shaft impact crusher or incremental damage such as in SAG mills). The fraction of the normal energy attributed to body breakage (single impact or incremental damage) is simply the dot product $\boldsymbol{D} \cdot \boldsymbol{N}$. The fraction assigned to chipping is then $(1.0 - \boldsymbol{D} \cdot \boldsymbol{N})$.

The mass loss rate due to abrasion and rounding of each particle in the simulation is proportional to the shear energy dissipation. Note that there is no minimum threshold energy for collisions to generate surface damage from these mechanisms. The mass loss from the abrasion is used to reduce the diameter of the particle, while the energy for rounding is used to reduce the blockiness and aspect ratios of the particles.

The fraction of the attrition energy attributed to each of the shape change components with w_1 being for diameter, w_2 for blockiness, w_3 for intermediate aspect ratio and w_4 for small axis aspect ratio is chosen according to the metric:

$$w_1 = \alpha \tag{4}$$

$$w_2 = \beta\left(\frac{10}{12 - m} - 1\right) \tag{5}$$

$$w_3 = \gamma(A_{xy} - 1) \tag{6}$$

$$w_4 = \gamma(A_{xz} - 1) \tag{7}$$

where α, β and γ are user selectable weightings, m is the blockiness and A_{xy} and A_{xz} are the two aspect ratios of the particle. In Eq. (5) the form is chosen so that w_2 is 0.0 for spherical particles which have $m = 2$ and increases as the particle blockiness increases. In Eqs. (6) and (7) the weights are chosen to be zero when the particles have aspect ratios of unity and to increase with increasing aspect ratio.

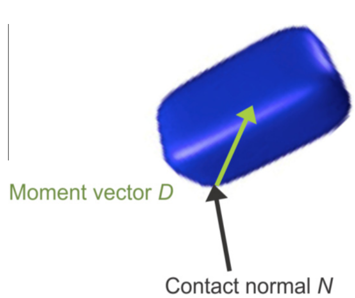

Fig. 2. Chipping and body breakage energy calculation using the contact normal N and the moment vector D from the contact point to the particle centre to apportion energy dissipation.

The weight vector $\boldsymbol{w}$ is then normalised to give the fractions of energy used to change each of the shape parameters of the particle from the energy dissipated in that collision.

Since the conditioning process is strongly preferential and abrasion is relatively small, we have used $\alpha = 1$ and $\beta = 10$ for the simulations reported in this paper. We use $\gamma = 5$ so that the sum of the energy used in the two aspect ratios is comparable to the energy used in the blockiness change. If a particle is spherical then $w_2 = w_3 = w_4 = 0$ and all the attrition energy is attributed to abrasion (and therefore goes into producing radial shrinkage of the particle). For a particle with the most extreme shape used in this paper having $m = 6$ and aspect ratios of 1.6 for both intermediate and small axes, the energy attribution weight vector is $\boldsymbol{w} = (0.05, 0.31, 0.31, 0.31)$. This means that all three shape attributes absorb the same fraction of the energy and the abrasion mechanism uses around 1/6 of this level. The energy used for changing each particle attribute is then converted to an explicit mass loss for the particle for that collision using a calibration constant. In this work, we use the same calibration value for each of the shape attribute calibration constants.

Once experimental data becomes available on the detailed shape variation of particles in some form of test mill then this data can be used to calculate suitable values for the breakage model parameters α, β and γ which are the calibration constants for each shape component. Here we use the indicative values noted above so as to demonstrate how the model works and the quantitative predictions of particle size and shape change that it enables. These mathematical forms used in Eqs. (5)–(7) are the simplest ones that capture the basic relationship between the energy absorbed and the geometry of the particle. When detailed experimental data becomes available for comparison this will allow refinement of these forms as well as calibration of the model parameters.

The mass loss rate due to chipping is chosen to be proportional to the fraction of the normal energy dissipation in the particle that is not directed towards the centre of the particle (as described in the last sub-section). It is controlled by the shape of the particle and the location of the contact. This energy component is the energy that determines the extent of the chipping of the corners of particles. This mechanism attempts to capture the type of breakage where an oblique impact near a sharp corner or edge is able to produce modest body breakage that removes the corner or edge in one or a small number of moderate sized fragments leading to a more rounded particle. Note that there is no minimum threshold energy applied for collisions to generate chipping damage. The chipping energy is attributed to the change of the blockiness and aspect ratios in the same way as was done for the rounding, except that the first component $w_1 = 0$ (as there is no diameter change produced by chipping).

Although this model is able, in principle, to predict mass loss from the parent particles due to chipping, rounding and attrition, it currently does not allow prediction of the fine progeny size distribution that results.

3.3. Modelling body breakage by incremental impact damage

The basic hypothesis of incremental breakage is that:

- At less than some specific energy (E_0), no body damage occurs as all deformation is elastic even though the particle surface may be abraded to some extent (Larbi-Bram et al., 2010).
- At higher energies, the particle accumulates incremental damage proportional to (E_i–E_0) and the probability of survival decreases with the level of accumulated damage until it breaks (where E_i is the energy absorption by the i-th collision of the particle).

The incremental damage model then has three components:

1. An estimate of the probability of breakage for each impact.
2. A rule for accumulation/severity of breakage.
3. A progeny distribution model (or breakage function) that allows the generation of a family of progeny for each breakage event.

Morrison et al. (2007) reported on various approaches to modelling incremental damage for this ore type. A satisfactory model for the probability of breakage, degree of breakage and likely progeny size distribution was developed based on the standard JKMRC impact breakage model (Napier-Munn et al., 1996) and the work of Vogel and Peukert (2004) with modifications suggested by Shi and Kojovic (2007).

In this model, the probability of survival after a series of k equal impacts is predicted by

$$S = 1 - \exp\{-f_{\mathrm{mat}} \cdot x \cdot k(W_{m,\mathrm{kin}} - W_{m,\mathrm{min}})\} \tag{8}$$

where

S = breakage fraction,
$W_{m,\mathrm{kin}}$ = single impact energy (J),
$W_{m,\mathrm{min}}$ = no damage energy (J),
x = diameter of particle (m),
k = number of (equal) impacts,
f_{mat} = material parameter.

The likely severity of breakage can be estimated in terms of the net energy absorbed by the body of the particle.

The Vogel and Peukert (2004) equation provides a useful model to test. In slightly modified form, the probability of selection S for breakage after i events

$$S = 1 - \exp\left(-b'\sum_i [E_i - E_0]\right) \tag{9}$$

where E_0 is the threshold energy per unit of particle mass (in J/kg) below which the particle does not accumulate any impact damage. Note that $[E_i–E_0]$ may not be less than zero and b' is a material parameter which may contain a particle size component. For more detail, see Morrison et al. (2006a).

The probability of survival after i events is simply $(1 - S)$.

$$P(\mathrm{Survival}/i) = \exp\left(-b'\sum_i [E_i - E_0]\right) \tag{10}$$

Whyte (2005) carried out a series of preliminary tests on 16×19 mm and 22×26 mm size fractions of a broad rock size distribution at four energy levels. These provide a manageable sub-set to test the survival model.

Breakage testing in this range of energies is unusual because the results of single impact testing appear to become erratic. For the early experiments, some preliminary testing was carried out to determine the energy level which would almost certainly cause breakage in a single impact. Once this level was established, separate sets of particles were subjected to successive impacts at fractions of this energy level. However, the progeny where more than a few impacts were required for breakage tend to be clustered around a t_{10} value of 5–6. Hence, it may well be that after the requirement for damage is exceeded; the cracks continue to propagate until internal friction equals the driving force suggesting a modified form for Eq. (10) where f is the fraction of E_0 at which the cracks will stop propagating.

$$P(\mathrm{Survival}/i) = \exp(-b'\sum_i [E_i - fE_0]) \tag{11}$$

The commonly used JKMRC relationship for severity of breakage can be modified in the same way following the strategy devised by Shi and Kojovic (2007):

$$t_{10} = A\left(1 - \exp\left(b'\sum_i [E_i - fE_0]\right)\right) \tag{12}$$

where t_{10} is the fraction of the mass of the original particle which will pass through an aperture of 1/10 of the original particle size – after the impact event. The analysis by Shi and Kojovic (2007) did not contain the factor f as almost all of their data was for single impact tests at much higher energies. This factor will only be important for multiple, low energy impacts. For high energy impacts, it will be absorbed into the fitted parameters of Eq. (10). The same parameter b' is used in both Eqs. (11) and (12) for this model. The degree to which this is correct is not currently fully clear. There can also be variation in b' with particle size. This was addressed at some length also by Shi and Kojovic (2007) but not all ores exhibit this variation. See Morrison et al. (2007) for further discussion of these issues.

Eq. (12) provides an estimate of the severity of breakage in terms of the t_{10} which will occur after multiple impacts which cumulatively lead to body breakage. It does imply a simplification as minor damage will probably also occur during each such event and some of this will be accounted for by the surface damage models. However, when the probability of breakage exceeds a suitable trigger energy level, a particle is assumed to break. This is of course a random distribution which matches the experimental distribution. The net accumulated energy provides an estimate of the actual t_{10} at which breakage occurred.

In a single impact event, t_{10} is predicted as a function of Ecs (that is, $E–E_0$). It seems reasonable to assume that the sum of the net crack propagation energy should generate a similar degree of new surface area. Therefore in the multi-event case, we expect each event which causes damage to continue to propagate cracks until some fraction – say f – of E_0 is reached. However, this raises the question as to whether the probability of actual breakage should also increase in proportion to the degree of damage which has been acquired to that time. Hence, it seems likely that the E_0 which we infer from multiple events may already contain this factor. Decoupling these effects would require an experiment which measures the actual stress as it is applied as well as measuring multiple particles to estimate the probability of actual breakage. This might require two parameters but it would be reasonable for the final stress to be a fraction f of the stress required to trigger crack extension. The authors are not aware of any published reports of such an experiment. However, this line of reasoning does justify the use of Eq. (12) with f set to unity to estimate the degree of breakage even though a proportionally greater stress than that producing E_0 may be required to trigger the damage event.

This incremental damage model is used in the DEM simulation to give predictions of body breakage due to damage accumulation. The energy E_i is amount of normal collision energy from the DEM collision model that is directed into the body of the particle (as described in Section 3.2 and recalling that energy is also directed into the surface erosion mechanisms as well). The probability for breakage to occur in the simulation is then determined from Eq. (9) and the resulting progeny from such a breakage event are predicted using the JKMRC breakage map (Narayanan and Whiten, 1988).

The size distribution of the progeny particles in the DEM simulation is calculated using a spline fitted version of Eq. (12):

$$t(k) = A(k)\left(1 - \exp\left(b(k)\sum_i [E_i - fE_0]\right)\right) \tag{13}$$

where $A(k)$ and $b(k)$ are coefficients for a set of $k = 6$ size classes from t_2 to t_{75}, which have been determined from drop weight tests. The method of Narayanan and Whiten (1988) can then be used to generate a full size distribution for the progeny. The method uses a matrix which relates other degrees of breakage in a particular event to the t_{10} parameter and then links those into a cumulative size distribution using a cubic spline function. It is well proven and in common use in comminution models (Napier-Munn et al., 1996). In terms of implementation in DEM, when a particle breakage event occurs, first the reduced size of the parent particle is calculated by extrapolation from the predicted $t(1)$ and $t(2)$ values. The model then iteratively determines the size of the next progeny particle by using the left over mass to calculate $t(\text{next})$ which is then substituted into the size distribution curve. The progeny are therefore estimated as a sequence of particles which exactly duplicates the splined, cumulative weight distribution (given in Eq. (13)) subject to an estimate of the largest particle in the progeny list based on $t(1)$ and $t(2)$.

The progeny model has been calibrated for breakage energies $E_0 > 0.3$ kW h/t. Use at lower breakage energies can cause the distribution curve to give physically unrealistic progeny sizes that are larger than the total remaining mass that is available for this and smaller progeny. In such cases, an estimate of the reduced parent size can still be determined from interpolation from $t(1)$ and the original parent size.

Eq. (13) is a general form of breakage behaviour that applies broadly to mineral rocks. The ore specific behaviour is determined by the parameters $A(k)$, $b(k)$, and E_0 which need to be measured or estimated for each ore type. Note, that in the DEM simulations it is usual to specify a minimum size of progeny fragment which will be resolved within the simulation. However, since the full empirical size distribution for each breakage is known, the finer unresolved progeny is known and can also be accumulated. The summation of these unresolved fragment distributions then gives a direct prediction from the DEM of the finer breakage products generated by the mill. In this way, the full size distribution for both the particles which are resolved in the simulation, and also the unresolved fine material can be predicted.

An important question about incremental damage that is not fully resolved is about what fraction of the incremental damage done to a particle leading up to its body breakage remains as damage inherited by its progeny. There is some evidence that a substantial fraction of the damage is used up in the fracture (Whyte, 2005). Whyte's thesis implicitly assumes that each particle's initial condition is "undamaged" and does not depend on its history. If it last passed through a crusher and was broken in a single event, then this is reasonable. A sample from a SAG mill load might however carry some history. On this basis, we set the initial damage energy for a new progeny particle to be zero in the simulations reported in this paper. It is quite possible though that there is some residual damage that should be quantified and inherited and it would be far from surprising if this was material dependent.

3.4. Replacement methodology in DEM for body breakage

The original body breakage method was proposed in Cleary (2001a). This used a replacement strategy where compressive or impact breakage fracture of a parent particle into finer progeny was represented directly in the DEM model. The size of the progeny and their packing arrangement into the body of the parent was determined by building an Apollonian packing of the parent. This model was limited by the use of spherical parent and progeny and the limited ability to include characterised breakage properties. Despite these limitations it has been able to provide useful predictions of crusher performance such as the VSI (vertical shaft impactor) and jaw crusher (Cleary, 2009), cone crusher (Cleary et al., 2013) and double roll crusher (Weerasekara et al., 2013), in a tumbling mill (Cleary, 2001a) and for a broad range of crushers in Cleary and Sinnott (2015) and Sinnott and Cleary (2015).

A more sophisticated variant that can represent breakage of non-round particles into non-round progeny was proposed by Delaney et al. (2010a). This was specifically developed to be able to model incremental breakage due to damage accumulation of particles over many collisions and uses breakage characterisation data from tests such as the JKDWT (JK Drop Weight Test) or JKRBT (JK Rotary Breakage Tester). The resulting set of progeny particles (from Eq. (13)) are packed into the volume of the parent particle. Details of packing of super-quadric particles and methods for densely packing these can be found in Delaney and Cleary (2010b). The progeny generation, packing and replacement of the parent are performed in the DEM simulation as an instantaneous event since the timescale for a particle to break is substantially shorter than any of the flow related timescales. The fraction of the mass of the fracturing parent particle that is in the progeny generated is principally controlled by the choice of the minimum fragment size. Packing fractions as high as 97% can be achieved when the minimum resolution size is very small compared to the size of the fracturing particle. More commonly a larger fragment size limit is used and resolved mass fractions are then more typically in the range of 70–90%. It should be noted that this is not a restriction on the DEM breakage model but rather a reflection of the usual DEM requirement to restrict the overall number of particles in a simulation so that the compute time is reasonable. Progeny that are smaller than the resolved size limit are independently accounted for which provides a prediction of the finer size fractions of the product material. Each breakage event, its progeny and the packing into the parent are performed on the fly during the DEM simulation. This allows good statistical sampling of the progeny size distributions. Use of pre-computed progeny and packing would lead to discretisation artefacts in the resolved progeny size distribution that would degrade the predictive capability of the breakage model. This approach has been used recently for predicting breakage in the Loveday mill (Delaney et al., 2013) and performance of a cone crusher (Delaney et al., 2015).

4. Experimental mill, ore characterisation and DEM setup

A substantial sample of a uniform, fine grained ore was collected in conjunction with detailed surveys of an industrial SAG operation. The sample was characterised using JKDWT abrasion and Bond ball work indices. Further repetitive DW test work was carried out at much lower energies to investigate incremental breakage of this ore (Morrison et al., 2007; Whyte, 2005). Bbosa (2007) found that E_0 ranged from 0.001 to 0.0015 kW h/t (3.6 to 5.4 J/kg) for smoothed particles or about one tenth of the Whyte estimate of 0.008 kW h/t for unsmoothed particles.

A sample of the ore was also tested in a 1.12 m by 0.31 m long semi-batch pilot mill. Key operating parameters for this mill are provided in Table 1. This mill was developed by Loveday and Hinde (2002) to calibrate and test an abrasion based AG/SAG model (Loveday and Whiten, 2004). This mill has been used because it provides for rapid removal of progeny as they are generated via slots at the base of the front of each lifter. This allows fine product to be rapidly removed minimising re-breakage and giving an attrition mass loss of fines per unit time. Additionally, the experiments were performed in a way that the mass of each particle in the charge was carefully tracked through time. This is ideal for detailed comparison with DEM predictions of the comminution process where the evolution behaviour of each particle can be explored. In addition to the mass of surviving particles, the progeny resulting from all the modes of comminution were collected, weighed and sized. This provides a mass balance around the mill.

Table 1
Key operating parameters of the pilot mill.

Mill diameter	1.12 m
Mill length	0.31 m
Lifters	14 – 40 × 40 mm
Mill speed	30 rpm
Test 1	30% autogenous load
Test 2	20% autogenous load
Test 3	10% – 10% SAG rock – ball
Rock charge	0.5–5.0 kg per rock at an SG of 2.65
Ball charge	80 mm nominal at an SG of 7.85

Tests were performed for 12 min for three different charges:

1. 30% AG charge,
2. 20% AG charge,
3. SAG charge with 10% rocks and 10% balls.

DEM simulations were run exactly matching these conditions. The simulated mill charge contains 53 rock particles for Test 1, 46 rock particles for Test 2 and 24 rocks and 45 steel balls for Test 3. In each test case, the exact starting mass of each rock is known and these were used in the DEM setup to create particles that exactly matched these. The rock particles used in the simulation were non-round which is required since the real rocks were not round. DEM should only be used to model particles as spheres if the particles are genuinely close to spherical (Cleary, 2004, 2009). To achieve this, the DEM particles were represented as super-quadrics (see Cleary, 2004 for details). The shapes of the particles in the experiments were not measured so these could not be used to specify the DEM particle shape. Instead, representative distributions of the shape factors corresponding to a hard ROM (run of mine) ore were used. The blockiness of the particles used (from Eq. (3)) was from $m = 3.0$ to 6.0 which produces particles from reasonably rounded to quite angular. For Tests 1 and 2 the rocks had an aspect ratio range from 1.0 to 1.4 for the intermediate axis (to major axis) and 1.1 to 1.65 for the minor axis (to major axis) aspect ratios. For Test 3 a more platey rock was used with intermediate aspect ratios of 1.0 and a minor axis aspect ratio of 1.0–2.0. Examples of such super-quadric particles are given in Fig. 1. The steel balls were almost spherical and so were represented as particles that were nearly spherical. The collision properties and numerical parameters used in the DEM model are given in Table 2.

Particles are divided into size classes with classes 1–10 being for balls and classes 11–20 being for rocks. The ball classes are only present in SAG case 3. The ball classes are uniformly distributed 74.65 mm and 81.44 mm. Since this range of variation is not large we will not report in detail about the specific ball size classes. The rock classes 11–20 are linearly distributed. The minimum size for the smallest rock class is 40 mm for all three test cases. The upper limit is 190 mm for Test 1, 169 mm for Test 2 and 183 mm for Test 3.

The collision model used in this work is a linear spring–dashpot model. Details of this and other collision models can be found in Thornton et al. (2011, 2013) for elastic and inelastic collisions. Many examples of simulation by the code can be found in Cleary (2004, 2009).

Table 2
Parameters used for the DEM simulations.

Normal spring stiffness	500,000 N/m
Shear spring stiffness	250,000 N/m
Density of rock	2650 kg/m^3
Density of ball	7850 kg/m^3
Coefficient of restitution (rock–rock)	0.3
Coefficient of restitution (rock–steel)	0.5
Coefficient of restitution (steel–steel)	0.8
Friction (rock–rock, rock–steel and steel–steel)	0.5

5. Size reduction from attrition, chipping and rounding

In the previous work (Morrison et al., 2006a), the standard JKMRC abrasion test mill (300 mm in diameter and 300 mm in length) was used for the calibration of the surface attrition rate constant. The abrasion mill was expected to be dominated by shear, so the shear energy absorption by the particles was used to calibrate the degree of abrasive wear expected in the 1.2 m diameter mill in terms of mass lost per unit of energy dissipated. In this paper, we take an alternative approach of using one specific particle from Test 1 to calibrate the surface mass loss rates against the DEM rate of shear energy absorption by the particles. In the experiments some specific particles were painted and each was carefully re-measured every 2 min so the specific mass loss rate of this and selected other particles is known to good accuracy. Most importantly, the correct matching of these particles at start and end could be made so that a reliable mass loss rate was experimentally known for these particles. The largest of these identified particles was used here to calculate the surface erosion calibration constants. The resulting mass loss constant was 0.07 g/J (which corresponds to an energy input of 4 kW h/tonne over the 10 min of the experiment, see Morrison et al., 2006b). This is ratio of the mass lost by the largest particle in the experiment with the amount of shear energy that it was predicted to absorb in the DEM simulation. This calibration constant was then used for the rock erosion rates in the simulations for all three tests (since the same rock was used in each of the experiments).

The same erosion calibration constant is used for all three surface erosion mechanisms since we have no current data available to enable a different choice to be made for the different mechanisms. The faster early rate of conditioning (rounding) of the rock particles observed in the experimental work in this pilot mill (Yahyaei et al., 2014) suggests that the rounding and chipping mechanisms do remove larger masses of rock per unit energy input than is found later for the abrasion. The corners and edges of the particles are intrinsically weaker than the middle of the particle so this type of differentiation would also seem intuitively reasonable. However establishing this properly would require access to more information on the masses of the rock particles at intermediate times throughout the milling process rather than just initial and final values that are used in this study. This question will be considered in future work.

5.1. Test 1: 30% AG load

Fig. 3 shows the size and shape evolution of the rock particles due to surface erosion by the abrasion, chipping and rounding mechanisms for AG Test 1. Initially, the rocks are quite angular and have reasonable aspect ratios. Collisions occur preferentially with corners and frictional contacts also preferentially occur at corners and edges. This concentrates the erosion at these more exposed locations leading to preferential rounding of corners and edges. As a result the particles become increasing round and increasingly equi-axed as well as smaller in diameter. In Fig. 3 the particles are coloured by their blockiness (the power m in Eq. (3)). The progress of the rounding can be seen as the particles become increasingly green and then blue. By 10 min of grinding, they are well conditioned and well rounded. Note that as they wear the particles better engage with the lifters and are thrown more efficiently. So by 10 min the centre of the charge region has few particles. Rather the particles are transported along the shell by

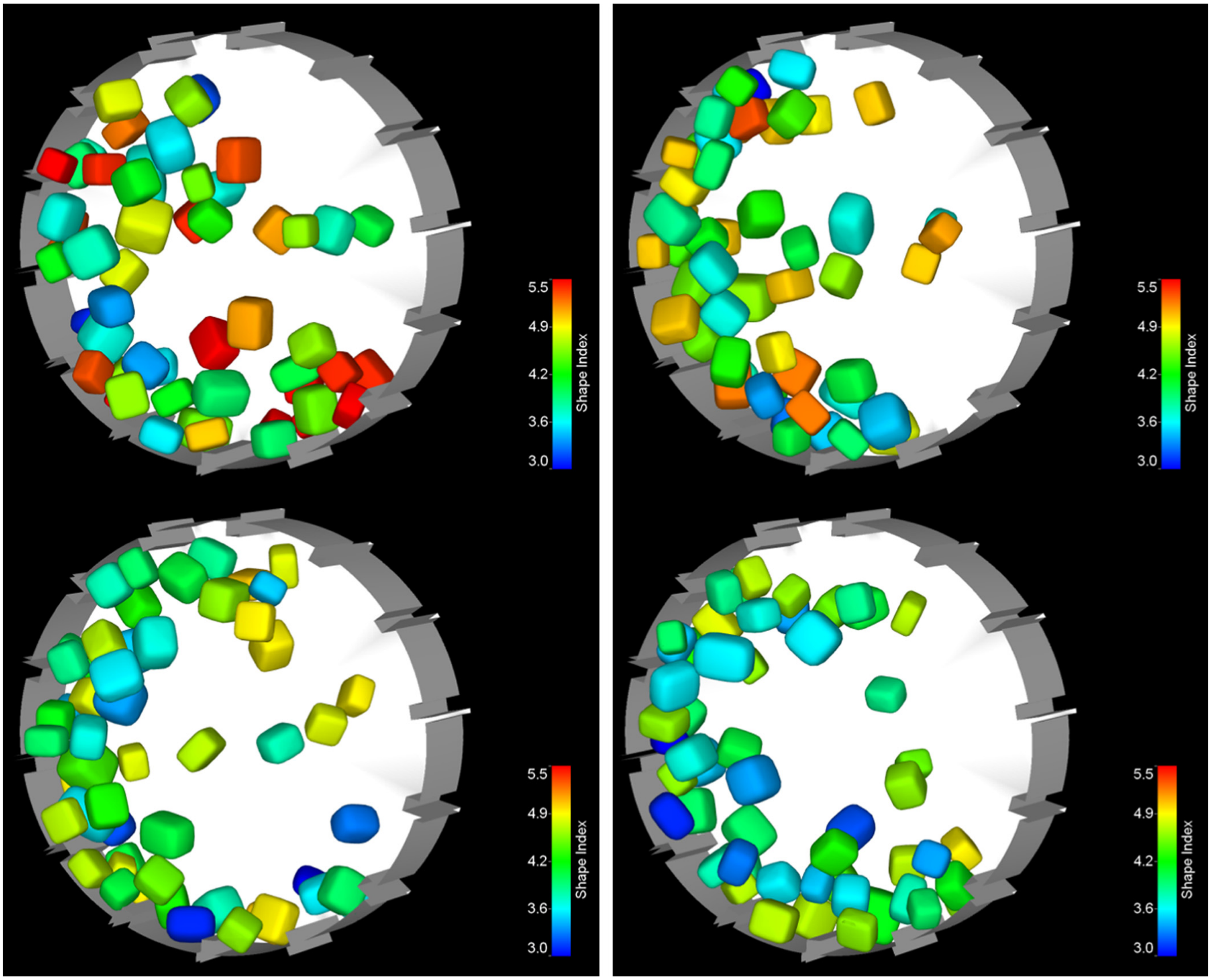

Fig. 3. Rounding and size reduction of rock particles in the pilot mill for a 30% AG charge for Test 1. The colours show the blockiness of the particles with red being quite angular and blue quite rounded with green being intermediate. The frames show the charge at 3, 6, 9 and 12 min. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the lifters in the lower left part of the mill (as seen in this figure) and then thrown as a relatively coherent stream from the shoulder on parabolic trajectories back down to the liner at the impact toe.

The predictions of the final masses of the charge particles for the autogenous Test 1 and the comparison to the experimental results are shown in Fig. 4. Only a few of the particles have not had some form of body breakage. These measured masses of these three particles are close to the line of particles masses predicted from the DEM. Recalling that the incremental breakage is not yet active in this simulation, the requirement is that the DEM predictions match or are larger than the experimental masses – which is achieved by the model in this simulation. For many of the particles this leaves around 40–60% of the total mass loss to be performed by the incremental damage model. This contrasts with the last increment of this work (Morrison et al., 2011) where all the damage was generated by the incremental damage mechanism but at the cost of using a lower value of E_0. In this new work, the reduced reliance on the incremental damage should enable use of larger and more physically reasonable E_0 values.

5.2. Test 2: 20% AG load

Fig. 5 shows the size and shape evolution of the rock particles due to surface erosion by the abrasion, chipping and rounding mechanisms for AG Test 2. The flow pattern and the particle size reduction predicted are similar to that found for Test 1 but with the charge being a bit more dilute because of the lower fill level used. The nature of the rounding of the particle shapes and the shrinkage of the particles due to the surface erosion mechanisms is similar to the previous case, indicating that the particle evolution dynamics are not particularly sensitive to the fill level at least in this pilot mill.

The predictions of the final masses of the charge particles for the autogenous Test 2 and comparison to the experimental results are shown in Fig. 6. Again, only a few of the particles have not had some form of body breakage. The line of final simulation masses again forms an upper limit on the experimental masses with two particles matching extremely closely. There is generally closer agreement for the masses for many of the particles for this case than there was for Test 1. This indicates that the incremental

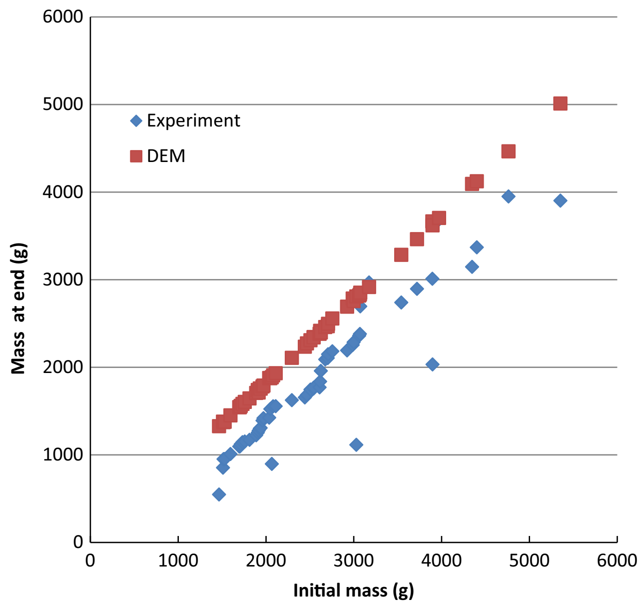

Fig. 4. Comparison of the rock size distribution for the experiment and the simulation for Test 1 with 30% AG charge after 12 min of grinding.

damage mechanism is less significant for the lower fill level with it needing to contribute only about half of the mass loss.

5.3. Test 3: 10% rock and 10% ball SAG load

Fig. 7 shows the size and shape evolution of the rock particles due to surface erosion by the abrasion, chipping and rounding mechanisms for SAG Test 3. The round grey particles are the steel balls. The balls tend to move on higher trajectories than the rocks because their smaller size and regular shape allows them to better engage with the lifters and so be thrown more efficiently than the non-round rock which tends to more occupy the middle region of the charge. The rate of mass loss is increased by the presence of the balls. The high circular arcing flow, particularly later in the simulation is more strongly established than for the AG charges with the balls being typically on the highest trajectories on the top side of the cataracting stream.

The predictions of the final masses of the charge particles for the SAG Test 3 and comparison to the experimental results are shown in Fig. 8. There is very good agreement for the larger rocks, which are too large for the impacts they experience to exceed their E_0 values and so they are not able to accumulate incremental damage. Consequently, their size reduction arises purely from the surface erosion mechanisms and the DEM model (using only the three erosion mechanisms) is therefore able to quite accurately predict the mass reductions of these larger particles. The surface interactions with the steel balls are likely to be at a higher average energy producing slightly better agreement for the SAG case than for the previous AG cases.

For particles of initial mass of around 2.5 kg and below, the incremental damage produced by the steel balls in the experiment is significant and there is a strong and systematic deviation of the experimental masses from the simulation mass line.

6. Predicted particle shape change from chipping and rounding

It is instructive to explore the nature of the shape change arising from the chipping and rounding mechanisms. Figs. 3, 5 and 7 give a qualitative idea of how the particles change but with DEM we are also able to explicitly quantify the nature of the changes. Fig. 9 shows the change in average values of the particle shape attributes for AG case 1 using the abrasion and rounding mechanisms. Fig. 9a shows the variation in blockiness over 12 min of milling. The rate of change in blockiness is higher in the early stages of milling and then slows between 4 and 8 min. It then decreases more slowly but linearly for the remaining milling time. The rate of change of blockiness in the later stages is 40% lower than in the early stages. This reflects the changing distribution of energy dissipation between mechanisms as the particles become rounder. Recalling that the energy used for chipping and rounding are the energy components directed away from the centre of the particle this means that as the particles become rounder the fraction of the energy being attributed to these mechanisms declines and the rate of chipping and rounding declines. This behaviour is to be expected and the ability of the model to predict such changes is encouraging. Fig. 9b and c shows the change in the average intermediate and minor axis aspect ratio. Both vary in a close to linear manner and unlike for blockiness there is no change in the rates of change over time. This suggests that the blockiness of the particles changes faster in response to these mechanisms than do the aspect ratios.

Recalling that the rocks are represented as super-quadrics with their blockiness uniformly distributed in the range $3 < m < 6$, the intermediate aspect ratio in the range $1.0 < A_{int} < 1.4$ and the minor aspect ratio in the range $1.1 < A_{minor} < 1.65$. We choose to use a similar approach for plotting the shape attributes as used for the size distribution, that is, the cumulative probability of passing a given value of the shape. We do this both because it is a familiar approach for representing size distributions and because the frequency distributions of the shape attributes are intrinsically noisy because of the very small numbers of particles in these test cases. Fig. 10 shows the cumulative distribution of the blockiness for AG case 1 at the start of the simulation and after 12 min of shape evolution caused by the surface erosion mechanisms. The initial distribution is shown in blue with diamond glyphs at the data points. There are no particles initially with $m < 3.0$ so the cumulative distribution starts here and then rising roughly linearly reaching 100% at $m = 5.75$ which is consistent with the specified range of the blockiness. The probability distribution used to generate the specific blockiness values for the particles is uniform – which is the reason for the broadly linear trend in the distribution. The bumpiness arises from the variation resulting from the small sample size of only 53 particles. After 12 min of milling the blockiness curve (shown by the red[1] curve with square glyphs at the data points) has moved to the left meaning that the particles have become rounder. The amount of movement is small for smaller values of m as there particles are already reasonably rounded and so less of the collision energy is directed into the rounding and chipping mechanisms. The change at the large value end of the distribution is substantial with the most blocky particle have a blockiness of only 4.5. The strong change in gradient of the distribution demonstrates the strong shape dependence of the chipping and rounding mechanisms. It is also worth noting that the distribution is now quite flat with little sign of variability arising from the small sample size. The milling has reduced the initial variability of the particle shape. This movement of the blockiness distribution is similar to the behaviour of the particle size distribution as the particles become finer with the distribution moving to the left – but in this case the movement reflects rounding rather than size change.

Fig. 11 shows the matching cumulative distribution for each of the particle aspect ratios for AG case 1, again at the start of the simulation and after 12 min of shape evolution. The initial distribution

[1] For interpretation of colour in Figs. 10 and 11, the reader is referred to the web version of this article.

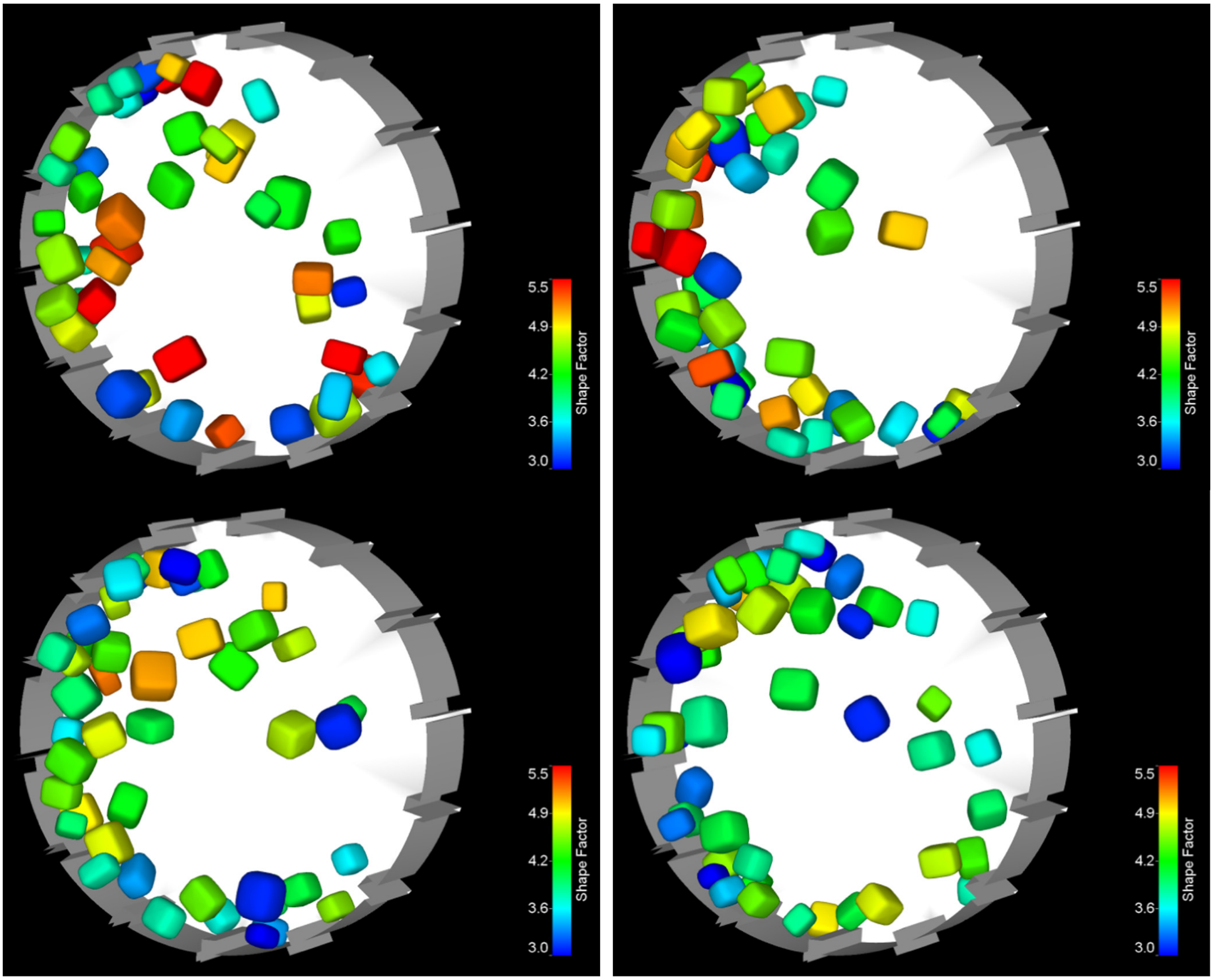

Fig. 5. Rounding and size reduction of rock particles in the pilot mill for a 20% AG charge (Test 2). The colours show the blockiness of the particles with red being quite angular and blue quite rounded with green being intermediate. The frames show the charge at 3, 6, 9 and 12 min. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of the intermediate axis aspect ratio (Fig. 11a) is quite smooth and starts with some particles having values of unity (meaning they are around the same width in the major and intermediate directions). All particles initially have intermediate aspect ratios of less than 1.4. After milling for 12 min the aspect ratios have all been reduced (as shown by the red curve being uniformly to the left of the original red curve) but the degree of change is much smaller than the blockiness. The gradient of the distribution has only increased slightly with only weaker preferential reduction in this aspect ratio being observed for larger initial values. The effect is significantly weaker than for the blockiness. The minor axis aspect ratio (shown in Fig. 11b) is similar in its behaviour. The initial distribution starts are value 1.1 and is below 1.65. It is again fairly smooth and linear. The degree of minor aspect ratio reduction from the chipping and rounding is larger for large aspect ratios and quite small for smaller values. The overall change in this aspect ratio distribution is arguably smaller than for the intermediate aspect ratio and is more concentrated in the larger part of the distribution but still much less so than for the blockiness.

Fig. 12 shows the cumulative distribution of the blockiness for AG case 2 at the start of the simulation and after 12 min of shape evolution caused by the surface erosion mechanisms. The change with milling observed is very similar in nature to that observed for case 1 but perhaps slightly weaker in the total amount of change. This indicates that the shape change arising from chipping and rounding is not particularly sensitive to fill level, at least for this scale of pilot mill. The behaviour of the aspect ratios is similar to that for AG case 1 (Fig. 11) and so is not shown here.

Fig. 13 shows the cumulative distribution of the blockiness for SAG case 3 at the start of the simulation and after 12 min of shape evolution caused by the surface erosion mechanisms. In this case there are round balls which do not evolve in shape and initially platey super-quadric rocks with blockiness in the range $3 < m < 6$ and minor axis aspect ratio in the range $1.0 < A_{minor} < 2.0$. The change in the blockiness distribution is stronger than for the AG cases with the final distribution being steeper than previously. There is also much more change in the moderately rounded sizes (around $m = 3$). The final distribution is noisier than for the earlier cases because there are only 24 rocks in this case. The behaviour of the aspect ratio is again similar to that for AG case 1 (Fig. 11) and so is also not shown here.

Summarising the effect of chipping and rounding mechanisms on the shape distributions of the particle attributes for all the cases, we find that the:

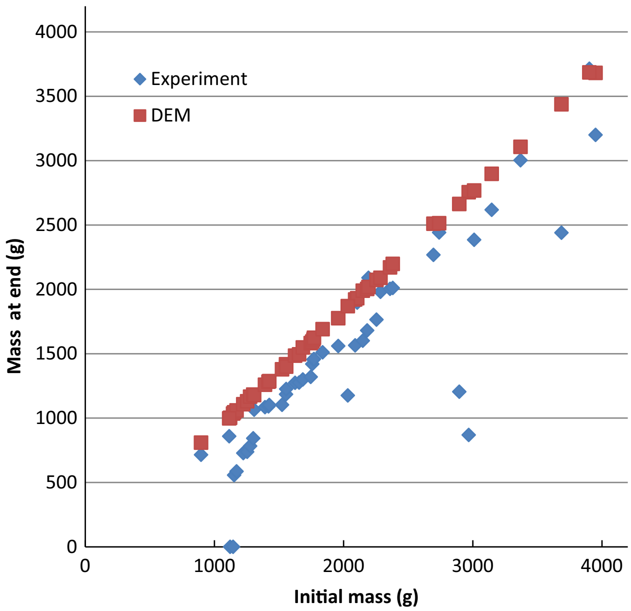

Fig. 6. Comparison of the rock size distribution for the experiment and the simulation for Test 2 with 20% AG charge after 12 min of grinding.

- Blocky or angular particles (those with $m > 4.5$) lose their corners and edges more rapidly compared to initially more rounded particles
- The cumulative blockiness distribution moves to smaller blockiness values and its gradient increases (meaning that the range of blockiness is much narrower)
- Blockiness varies much more rapidly than aspect ratio.
- The rate of aspect ratio change depends on aspect ratio with particles of 1.2 or less changing this aspect ratio only very slowly but with faster reduction rates for higher aspect ratios.
- Changes in the average shape properties are almost linear – essentially proportional to the mass loss rate. However, the average blockiness decline is faster in the early stages but then decreases to be around 40% lower in the latter half of the milling time.

7. Change in energy utilisation as rock particles erode

In the physical experiments the amount of rock mass loss by attrition is moderate. In order to better demonstrate the effect of rock erosion on the pattern of energy utilisation, we increase the attrition rates by a factor of 10. This means that the mass losses are 10 times larger and the particles are significantly smaller at the end of grinding. We then compare the energy usage patterns for the first minute of operation (initial performance) and the final minute (highly eroded rocks) for both AG case 1 and SAG case 3 to see how the mass loss has affected the nature of the collisional environment and how it uses the energy supplied.

The initial power draw for AG case 1 is 791.2 W (calculated over the first minute of milling) for a charge mass of 137.6 kg. This declines to 380.7 W at the end of the milling process at 12 min due to the large reduction in the mass of the rocks (62.7 kg). This represents a 52% reduction in power draw for a 54% reduction in charge mass. This means that the specific power draw (ratio of power draw and charge mass), changes only slightly from 5.75 W/kg up to 6.07 W/kg for the heavily eroded rocks. In contrast SAG case 3 has a similar initial power draw of 850.4 W but a much larger (compared to the AG case) power draw of 580.9 W at the end of grinding. Although the rock has been heavily eroded (from 46.6 kg at the start down to 13.8 kg at the end) the large mass of balls (86.0 kg) has not changed and they continue to dissipate significant amounts of energy throughout the milling process. More quantitatively, there is a 31.7% reduction in power draw compared to a 24.7% decline in charge mass. This means that the specific power has decreased from 6.4 W/kg for fresh charge down to 5.82 W/kg at the end. Note that these values are similar to those found for the AG case, but the specific power draw decreases for the SAG case instead of the increase observed for the AG case. In broad terms though, the power draw is essentially proportional to the charge mass.

The rock–rock type of collision initially consumes 53% of the energy supplied by the mill for AG case 1. This declines to only 38% at the end of milling. The remaining energy is used in rock–liner collisions. This indicates that as the particles become smaller they are less likely to collide with each other and more likely to fall completely through the charge and impact directly onto the liner. In contrast, for the SAG case 3 there are five different combinations of collisions possible. The energy used in each type of collision is shown in Table 3. Initially the rock–rock collisions consume only 11% which is much lower than for the AG case. This is due to the small number of rocks and the relatively larger number of balls which dominate the collisions. As the rocks shrink, the direct rock–rock collision energy consumption drops drastically to only 1.9% of the energy consumption as the increasing small rocks able to collide with sharply decreasing frequency. The rock–liner collisions dissipate a larger fraction of the energy (around 18%) which is much smaller than the 47% occurring for the AG mode. This proportion decreases by 25% as the particles shrink. Since the dominant ball component of the charge does not change over time, the chance of the decreasing sized rocks hitting the liner directly is not significantly affected. The rock-media collisions initially dissipate a larger fraction of the energy at 21.7%. This declines by 38% with grinding reflecting the smaller rock masses and their lower chance of being hit by the media. The energy dissipated in media-rock collisions is initially 1.96 times higher than for rock–rock collisions. This increases to 7.1 times for the smaller rocks at the end of the grinding process. This highlights the importance of the grinding media and how large a role they play in size reduction and how this importance increases as the ball to rock mass ratio increases. The remaining energy consumption does not contribute to comminuting the rock particles, but simply leads to media and liner wear. The media-liner collisions initially consume a massive 36.1% of the energy rising to 48% for the worn charge. The direct media on media collisions initially consume 17.3% of the energy which increases strongly to 24% as the decreasingly small rock particles are less frequently trapped between approaching media. So the inclusion of media maintains a more constant grinding environment for the rock particles but at a cost of a significant increase in energy consumption (all of which is wasted in via media and liner wear). This extra energy consumption explains the entire observed difference in power draw between case 1 and case 3 at the end of the grinding process.

It is also instructive to examine the distribution of energy used for each of the size reduction mechanisms for each case at both the start and end of the grinding process. The start and end distributions of energy between mechanisms for AG case 1 are given in Table 4. The energy dissipated in boundary wear is substantial at around 24% at the start rising above 31% at the end (due the increased weighting of rock–liner vs. rock–rock collisions. Since the mass of rock is changing as is the fraction of energy absorbed by the rocks it is also useful to consider the relative contribution of each mechanism as a percentage of the total damage energy absorbed by just the rocks which is given in Table 5. This data is obtained by re-normalising the first 5 rows of data from Table 4.

Initially, the dominant rock damage mechanism is rounding with around 36% of rock damage energy being used to remove

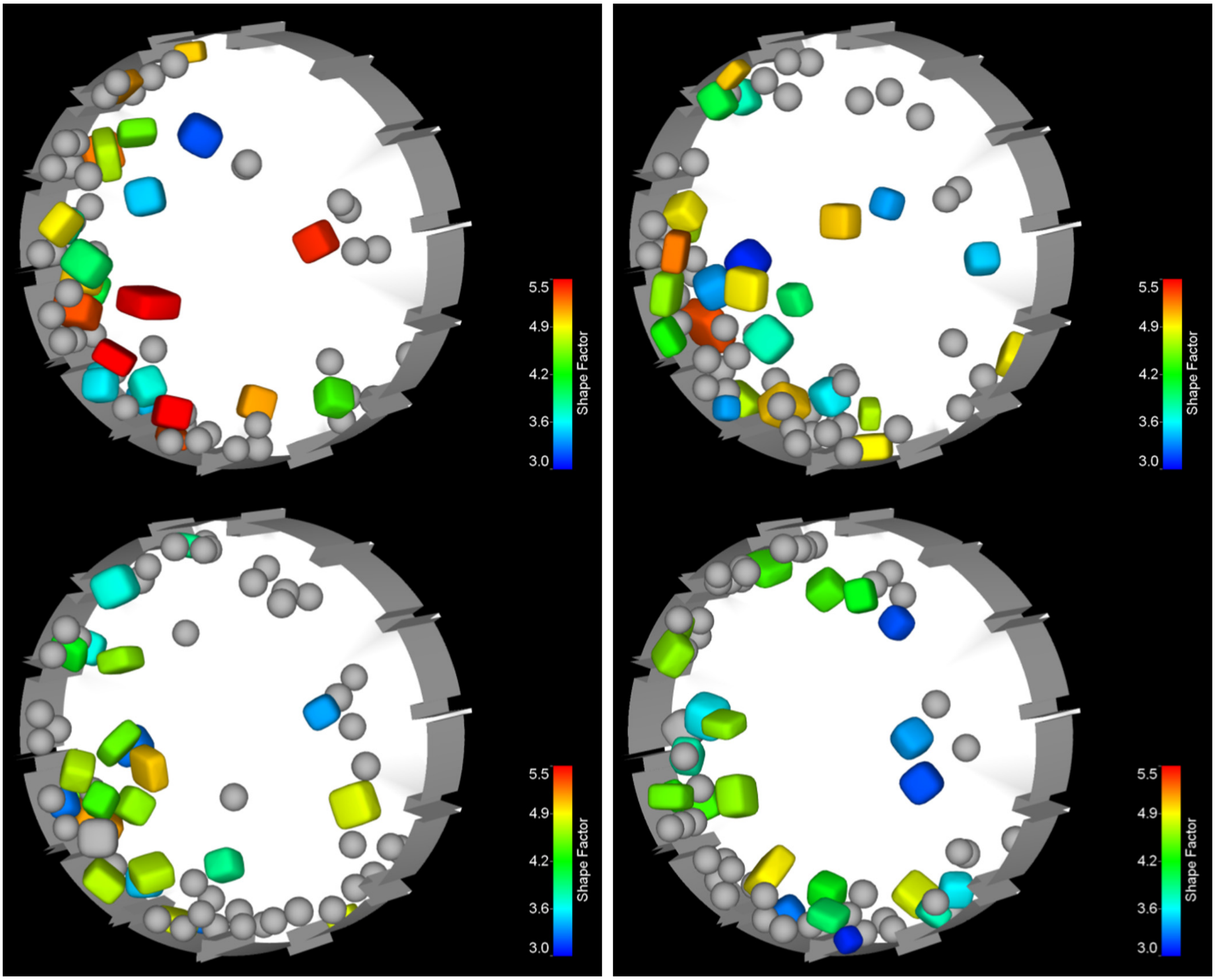

Fig. 7. Rounding and size reduction of blocky rock particles in the pilot mill for a 10% rock and 10% ball SAG charge (Test 3). The colours show the blockiness of the rock particles with red being quite angular and blue quite rounded with green being intermediate. The steel balls are round and shaded grey. The frames show the charge 3, 6, 9 and 12 min. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the corner material on the particles and thereby causing the strong shape change observed. The chipping mechanism at 21.6% is also quite strong and also contributes meaningfully to rock shape change. The energy available for incremental breakage (which is the part of the normal component energy dissipated that is directed towards the centre of the particle and which also exceeds the E_0 value – set here at 1.0 J/kg) is only 2.6% of the rock damage energy (and 2.0% of the total energy). This explains the low level of incremental damage observed in the experimental results for case 1 and the general difficulty of tumbling mills to break larger particles. The wasted energy is defined here as the total energy from the normal component of collision energy which is directed into the particle and which is less than the elastic threshold E_0 and which therefore does not contribute to volumetric damage of the particle. This is a substantial fraction of the rock damage energy at 31.6%. This energy is dissipated as heat within the particles. This energy wastage is one of the key reasons that tumbling mill performance is inherently inefficient.

With the identification of these mechanisms and their methods of calculation, we can for the first time estimate from simulation the fraction of energy wasted in a tumbling mill. For this case it is 51.4% of the total energy available. The mechanism with the smallest damage energy utilisation is abrasion which consumes only 6.6% of the damage energy (and 5% of the total energy). This is responsible for the radial shrinkage of particles due to frictional sliding. An important reason for this being so small a fraction relates to the particles initially being quite angular and having higher aspect ratios which leads to very high corner and edge erosion from rounding and chipping. Unfortunately, the abrasion mechanism is the one that is responsible for the shrinkage of large particles bringing them down to the size range at which the mill can start to produce body breakage, so a slow rate of abrasion means that it takes a long period for the particles to become small enough for this to happen. The incremental damage is the mechanism then responsible for this body breakage and this is only 2.6%. So the fraction of the energy being used in the two most important mechanisms for significantly reducing the size of rock particles is only 7%. Around 43% of the energy is used for removing corners and edges, a process often described as conditioning (Loveday and Naidoo, 1997). This does contribute to the creation of finer rock particles but these mechanisms also serve to protect the integrity of the rocks by preventing energy being absorbed by

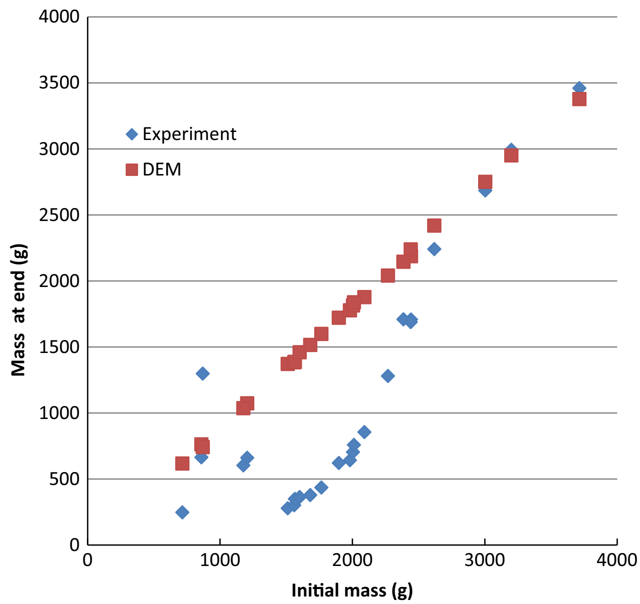

Fig. 8. Comparison of the rock size distribution for the experiment and the simulation for Test 3 with 30% SAG charge after 12 min of grinding.

the core of the particles (which is not at all what one would like to happen in a grinding mill in terms of the larger particles while not forgetting that production of finer particles is also important).

At the end of the grinding process (see the third column of Tables 4 and 5) the distribution of energy between mechanisms has changed substantially. The fraction of energy in the liner wear has increased strongly so the amount of energy being used for grinding is reduced. However, the mass of rock to grind has also declined. The fairest comparison is to consider the relative distribution of rock damage energy between the rock breakage mechanisms, as given in Table 5. The particles having become much more rounded causes more energy to be directed towards their centres. This leads to an 82% increase in abrasion to a still relatively small percentage of 12%. There is a sharp corresponding decrease in the amount of rounding (due to the particles now being much rounder). In contrast, the amount of damage energy used for chipping has increased strongly by 57%. This change in the balance of rounding and chipping (which both depend on particle shape) is a result of the changing balance of collisions being from rock–rock (which has a larger sliding frictional component to their interaction) to rock–liner (which is more predominantly in the normal direction). The strong rounding of the particles at the end of the grinding period means that a larger fraction of the increased energy in the normal direction is directed into the core of the particles causing more collisions to exceed the elastic threshold E_0. This leads to a strong 73% increase in the amount of incremental damage. For the wasted energy, the increased fraction of normal energy is counter-balanced by the decreased fraction of energy below the elastic threshold, which combined give a small decrease of 2%. It is worth noting that even with these large changes the incremental damage and abrasion mechanisms are still relatively weak (only 11.4% of energy consumption and 16.5% of the rock damage energy) compared to the other mechanisms.

Table 6 shows the distribution of energy use by the mechanisms for SAG case 3. The quantities in the first five rows are calculated for just the rock particles. An additional row gives the percentage of energy consumed by the media. This is dominant at the start at 41% and increases slightly to 42.9% at the end of grinding. The liner wear mechanism contribution for the SAG case is the second largest energy consumption mechanism, initially at 26.2% and

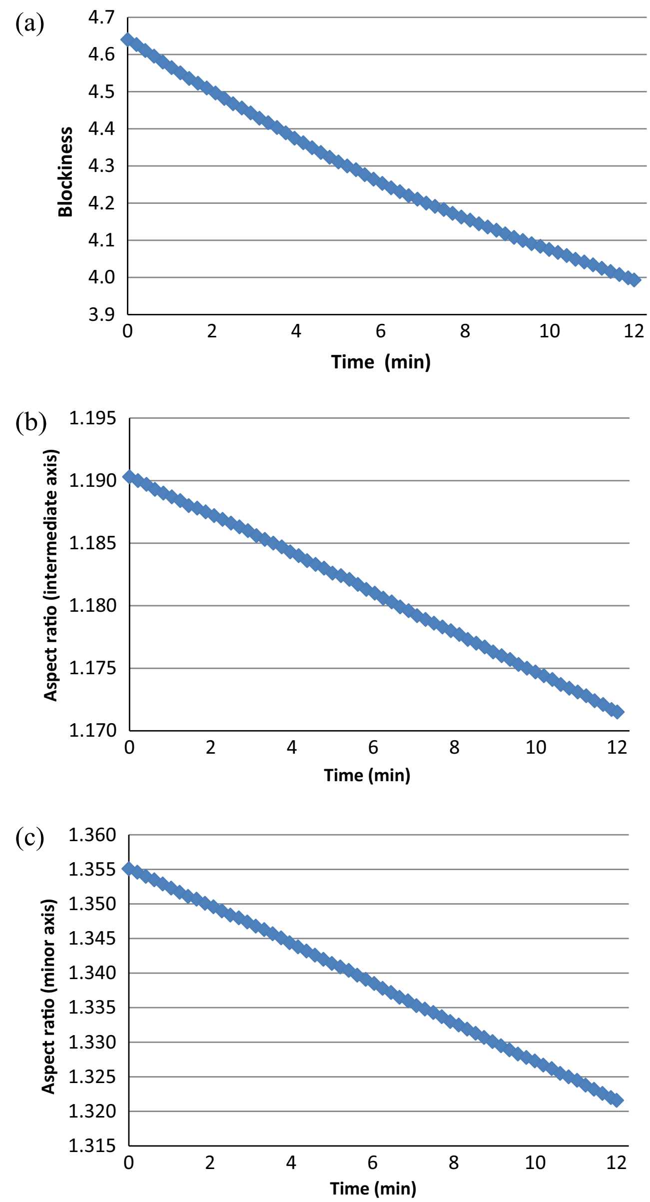

Fig. 9. Change of average shape attributes for AG case 1, (a) blockiness, (b) intermediate axis aspect ratio, and (c) minor axis aspect ratio.

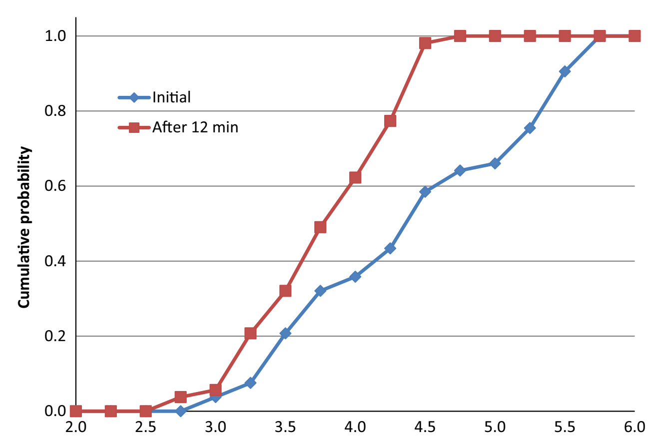

Fig. 10. Change in blockiness distribution as rock particle shape evolves for AG case 1.

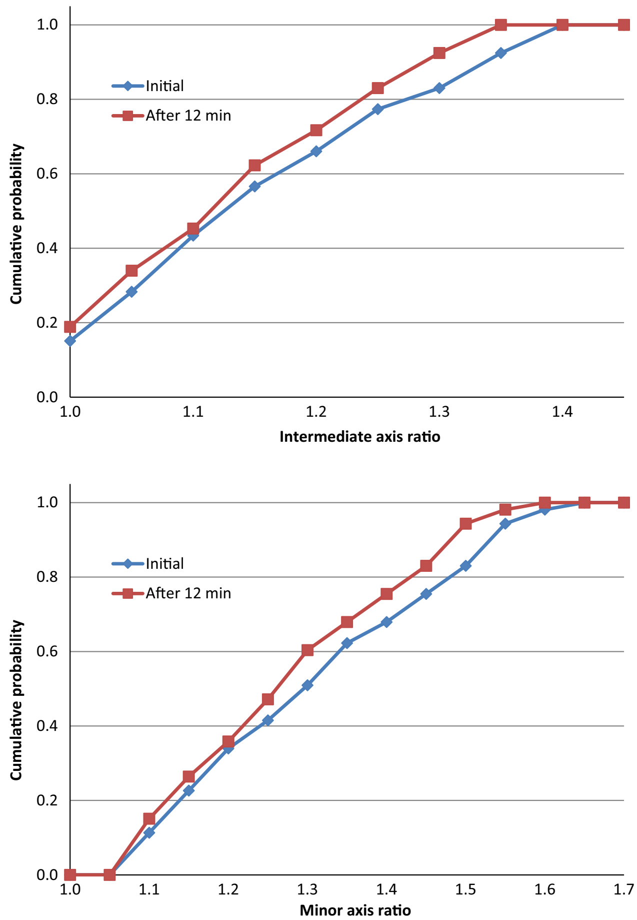

Fig. 11. Change in aspect ratio distributions of rock particles as their shape evolves for AG case 1.

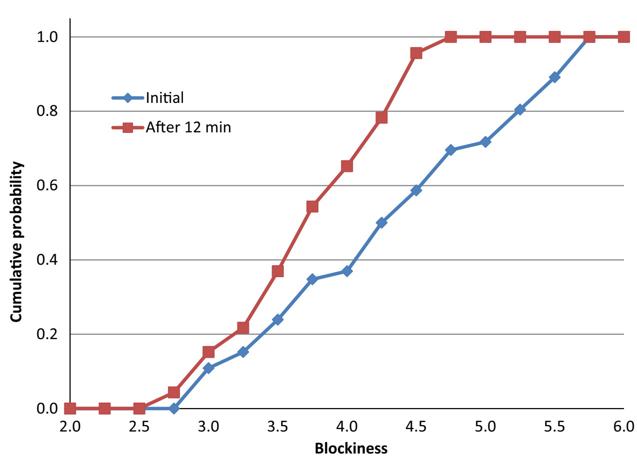

Fig. 12. Change in blockiness distribution as rock particle shape evolves for AG case 2.

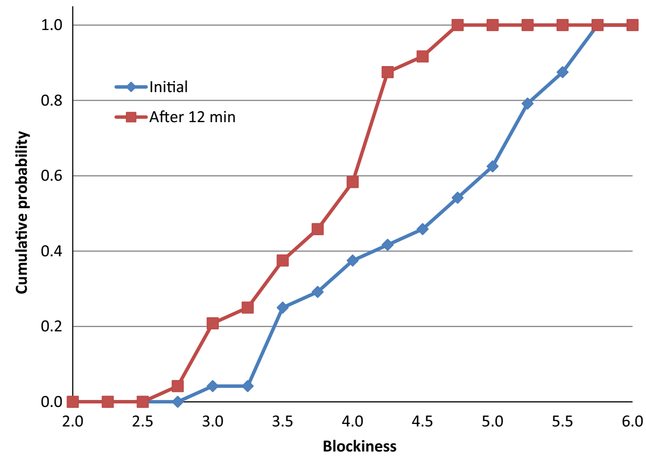

Fig. 13. Change in blockiness distribution as rock particle shape evolves for SAG case 3.

Table 3
Energy consumption for different collision types at the start and end of milling for SAG case 3.

Collision type	Initial percentage (first minute)	Final percentage (last minute)	% change over time
Rock–rock	11.1	1.9	−83
Rock–liner	19.2	12.4	−25
Rock–media	21.7	13.5	−38
Media–media	17.3	24.2	40
Media–liner	30.8	48.0	56

Table 4
Distribution of energy dissipation between different mechanisms at the start and end of the milling process for AG case 1.

Dissipation mechanism	Initial percentage (first minute)	Final percentage (last minute)	% change over time
Abrasion	5.0	8.3	66
Rounding	27.3	12.9	−53
Chipping	16.4	23.4	43
Incremental damage	2.0	3.1	55
Wasted	27.4	21.4	−22
Liner wear	24.0	31.1	30

Table 5
Relative contributions of the breakage mechanisms to the rock comminution at the start and end of the milling process for AG case 1.

Dissipation mechanism	Initial percentage (first minute)	Final percentage (last minute)	% change over time
Abrasion	6.6	12.0	82
Rounding	35.9	18.7	−48
Chipping	21.6	34.0	57
Incremental damage	2.6	4.5	73
Wasted	31.6	31.0	−2

28.9% at the end. This is slightly larger than for the AG case at the start and then slightly lower at the end. Between these two energy consumption mechanisms, around 47–52% of the energy input is used purely for wearing the liner and the media.

The fraction of the damage energy used by each mechanism to comminute the rocks for SAG case 3 is given in Table 7. This is a renormalization of the first five rows of the data in Table 6. At the start of the grinding process, the balance of damage mechanisms is very similar to that of the AG case. The largest difference is for the incremental damage which is relatively smaller at 1.7% compared to 2.6% for the AG case. At the end of the grinding process, the abrasion fraction of damage energy is only modestly increased, the rounding has declined by just 9%, the chipping by only 2% whilst the incremental damage has increased sharply by 88% and the wasted energy has increased slightly (by 8%). The changes observed over time for the SAG case are different in

Table 6
Distribution of energy dissipation between different mechanisms at the start and end of the milling process for SAG case 3.

Dissipation mechanism	Initial percentage (first minute)	Final percentage (last minute)	% change over time
Abrasion	2.1	2.1	0
Rounding	12.0	9.4	−22
Chipping	7.0	5.9	−16
Incremental damage	0.56	0.90	61
Wasted	10.7	10.0	−6
Media wear	41.0	42.9	5
Liner wear	26.2	28.9	10

Table 7
Relative contributions of the breakage mechanisms to the rock comminution at the start and end of the milling process for SAG case 3.

Dissipation mechanism	Initial percentage (first minute)	Final percentage (last minute)	% change over time
Abrasion	6.4	7.5	17
Rounding	36.6	33.2	−9
Chipping	21.3	20.8	−2
Incremental damage	1.7	3.2	88
Wasted	32.6	35.3	8

pattern and generally smaller in magnitude than for the AG case. The large increases in abrasion and chipping observed over time for the AG case do not occur for the SAG case and the chipping fraction is essentially constant instead of increasing. The only mechanism that increased more over time for the SAG case was the incremental damage, increasing modestly more at 88% compared to 73% for the AG case. Note that while it has increased more the fraction of incremental damage is still smaller for the SAG operation. The grinding media (which does not change size) appears to maintain a more constant grinding environment over time as the rock particles are comminuted. This suggests that full size AG mill performance could be somewhat more sensitive to the changing size and shape distributions of the rock axially along the mill from feed to discharge. In contrast, the presence of the media in a SAG mill appears likely to maintain a more constant grinding environment along the mill with the performance perhaps less influenced by the specific nature of the rock charge variation within the mill.

This is consistent with industrial experience that SAG mills are easier to operate (Napier-Munn et al., 1996). Hence, SAG mills dominate the industry. When AG milling was in more common use, some designers provided for storage of coarse and fine feed fractions to improve control. In practice, operators tended to use up the faster milling coarse feed and leave the next shift with a problem and no autogenous grinding media.

It is also worth briefly comparing the actual SAG environment with the test mill environment. While the test mill environment is depleted of fines and smaller particles compared with an industrial SAG mill, the implications for process (in)efficiency are still reasonable.

An interaction between a 60 mm and a 6 mm particle can break the smaller particle at a t_{10} of A (the largest degree of breakage possible in a single impact) without much effect on the stress level of the larger particle because it masses about 1000 times more. Similarly, the destruction of the 6 mm particle does not produce much product compared with breakage of the larger particle. Earlier DEM modelling of a 1.8 m $\times$ 0.6 m industrial pilot SAG mill (Cleary and Morrison, 2004), suggested that most interactions between particles occurred within the same size fraction or one above or below for a combination of geometric access and probability density reasons. That is, there have to be enough particles of similar sizes to interact with. The accepted ball mill type grinding model of a layer of slurry on a particle offers much less mass again to absorb any useful energy. By way of contrast, a glancing blow which shears off an asperity only needs to generate a small quantity of new surface area. A normal impact is more likely to load a particle and begin to damage it through induced intra-particle tension at stress levels with elastic energy greater than E_0. A gouging action is more plausible for incremental surface damage – once a particle has been sufficiently smoothed. This is consistent with the effective friction model tested by Djordjevic at al. (2006) and the results in this paper.

Musa and Morrison (2009) investigated tumbling mill efficiency by comparing measured energy with that required to achieve a similar product using a JK drop weight tester. SAG mill efficiency was found to be typically about 40%, that is, about two and a half times as much energy was required compared with single impact breakage. The incremental breakage hypothesis is sensibly consistent with this analysis.

8. Effect of size reduction on energy spectra

Energy spectra are constructed by calculating the energy dissipation in each separate collision and then building a probability distribution of the different collision energy levels. These can be constructed using either the collision rate or the energy dissipation rate as the dependent variable and the independent variable can be either collision energy (J) or specific collision energy (J/kg). These spectra are a way of characterising the collision environment in a mill and providing a quantitative framework for evaluating the effects of changes in design and operating variables. Since the collision between particles generate the breakage of the rock, then changes in the collision spectra link directly to changes in product. The form of the energy spectra used here was introduced by Cleary (2001b). It has been used for example to look at the effect of changing feed size to a coarse clinker grinding ball mill (Cleary et al. 2008) and for tower mills (Sinnott et al., 2006). Other authors have used variants of the spectra for similar purposes. For the current paper we will use the form from Cleary (2001b) and will use collision rate and specific collision energy as the dependent and independent tabulation. Spectra are generated for each size class, for each collision type and over all collisions. Here we will focus on the spectra for the specific size classes.

Figs. 14 and 15 show some energy spectra from AG case 1 and SAG case 3, respectively. Again the attrition rates have been increased by a factor 100 so that there is a large change in charge particle size. A number of important questions relating to the change in behaviour of the energy spectra with changing charge attributes can be explored with these simulation cases.

The first question is whether there is any difference between the spectra of the AG and SAG charge configurations. Comparing Fig. 14 (top) and Fig. 15 (bottom/left) for size class 16 at the start of the grinding process we observe that the spectra for this common size class is similar in shape and magnitude. There is a central model maximum which is indicated by the vertical line. The collision frequency declines slowly for both higher and lower energy. There is a long tail with energies recorded out to 10^{-8} J but these are of no importance since they do not contribute to breakage and are not shown. At higher energies the collision frequency declines at an increasing rate dropping sharply at energies above 1 J with peak energy observed of around 5 J. The collision frequencies are slightly lower for the SAG case because of the smaller number of particles in the charge to generate the collisions. The peak energy is similar at around 5 J and the modal peak is modestly higher at around 0.028 J/kg. It is important to note that the vast majority of the collisions are very weak. The E_0 value divides this

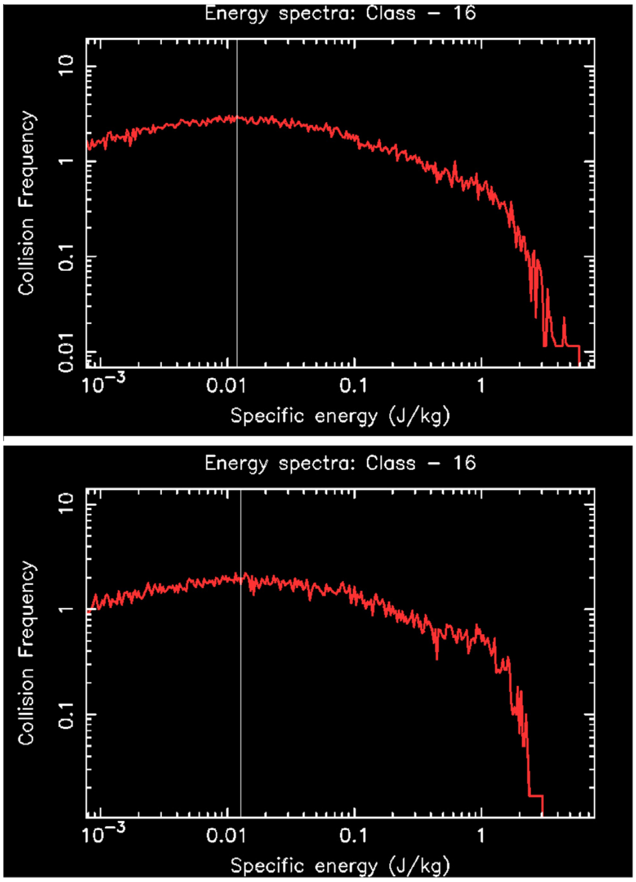

Fig. 14. Top end of the collision energy spectra for size class 16 (finest feed class) at start of milling process and after 12 min for AG case 1.

spectra into a small range of collisions with collision energies above this level that contribute to incremental damage and ones below this which do no damage to the particles and so contribute only to the wasted energy. However, if the particle is coated with slurry, the fine particles in the slurry may be comminuted within the contact zone as the energies required are most strongly dependant on mass. A 0.5 mm particle has a mass which is only 1/125,000 that of a 25 mm particle. Many of these fine particles may be broken by an impact which has negligible effect on the larger particle.

It is further important to note that the collision frequencies of the higher energy collisions are quite small (between 0.1 and 0.01 Hz). This means that collisions able to do damage to rock particles in this mill occur on average between once every 10 and 100 s. Over the duration of the grinding process this is only 7–70 collisions spread across the particles in the class, so around 1–10 collisions per particle. Only these small numbers of collisions are able to create any form of body damage to the rock particles. The small differences between the AG and SAG spectra cause one to ask the question as to how can there be such a large difference in the observed final size distribution for the SAG case (in Fig. 8) compared to the AG case (in Fig. 4)? This question will be explored in the next section.

The second question is about whether the energy spectra vary with time (i.e. with the degree of particle shrinkage) for the same size class? That is, does the shrinkage and shape change of the particles in the charge change the nature of the collision environment that a particle of the same size experiences. Fig. 14 shows the upper end of the collision energy spectra for size class 16 (finest feed class) at start of milling process and after 12 min for AG case 1. There is little difference in these spectra with the shape being similar, the modal peak being little changed and even the very sensitive peak collision energy varying only slightly. The energy absorbed per unit mass is similar for similar mass particles despite long milling and shape change. There is a subtle change in the shape of the spectra at collision energies of 1–2 J with secondary inflection having developed. There is a hint of this inflection point in the initial spectra but it has become more clearly defined at the end of the grinding process. This modest increase in collision frequency for this narrow range of energies does not have any clear cause nor does it have any obvious consequence. Note that the actual particles that are recorded as class 16 are not the same particles as those recorded in that class at the start. The original class 16 particles for AG case 1 have become class 14 in AG case 1 because of their size reduction from surface erosion. The particles in class 16 at the end of 12 min of grinding are typically those initially found in class 19.

The final (third) question is around how does the energy spectra vary with size? Here we use the spectra from SAG case 3. Some of the spectra (even numbered size classes) are shown in Fig. 15. The spectra for classes 12 and 14 (which are product size classes) are from the end of the simulation whereas classes 16 and 18 are feed classes and are taken from the start of the simulation. The spectra for class 18 (a larger feed class) is quite similar to that of the previously discussed class 16. For class 14, the upper half of the spectrum is similar but there is an increase in frequency of lower energy collisions. For class 12, the changes are similar in nature but more extreme leading to an almost constant collision frequency for collisions below 0.2 J. This indicates that the particles experience increasing numbers of weak collisions as they become smaller.

Table 8 shows the peak collision energy intensity for different rock size classes for AG case 1 and SAG case 3. Peak energies are taken to be the largest energy which has collisions with a frequency of 0.01 Hz or more. The spectra used are for the full 12 min of mill test operation so there are rock particles in both feed and product size fractions. The threshold frequency of 0.01 corresponds to around one collision per particle over the 12 min duration. Less frequent collisions can over no effect on the rock charge over this duration. There is a general trend of increase in the peak (most frequently observed) collision energy per unit mass as the particle size becomes smaller for both AG and SAG type charges. This means that as a particle shrinks the proportion of its energy spectra that exceeds the E_0 elastic threshold increases. Therefore, an increasing fraction of the available collision energy is able to manifest as incremental damage. So as a particle shrinks in size from abrasion, chipping and rounding its rate of accumulation of incremental damage increases and its chances of undergoing catastrophic body breakage increases. This highlights the complex interaction between the different breakage mechanisms that control the evolution life cycle of particles within a tumbling mill.

9. Why are ball collisions more effective at breakage than rock collisions?

A critical concern with the energy spectra results shown earlier is that there is little meaningful difference between those for AG and SAG modes. The peak energies are both around 3.9–4.6 J. On this basis, one would expect relatively similar amounts of incremental damage for cases 1 and 3. However, Figs. 2 and 6 show that there is clearly a significant contribution to size reduction for SAG case 3 for the smaller and intermediate size particles. This observed behaviour cannot be explained except if the SAG case energy spectra were to have a clearly higher peak value for its

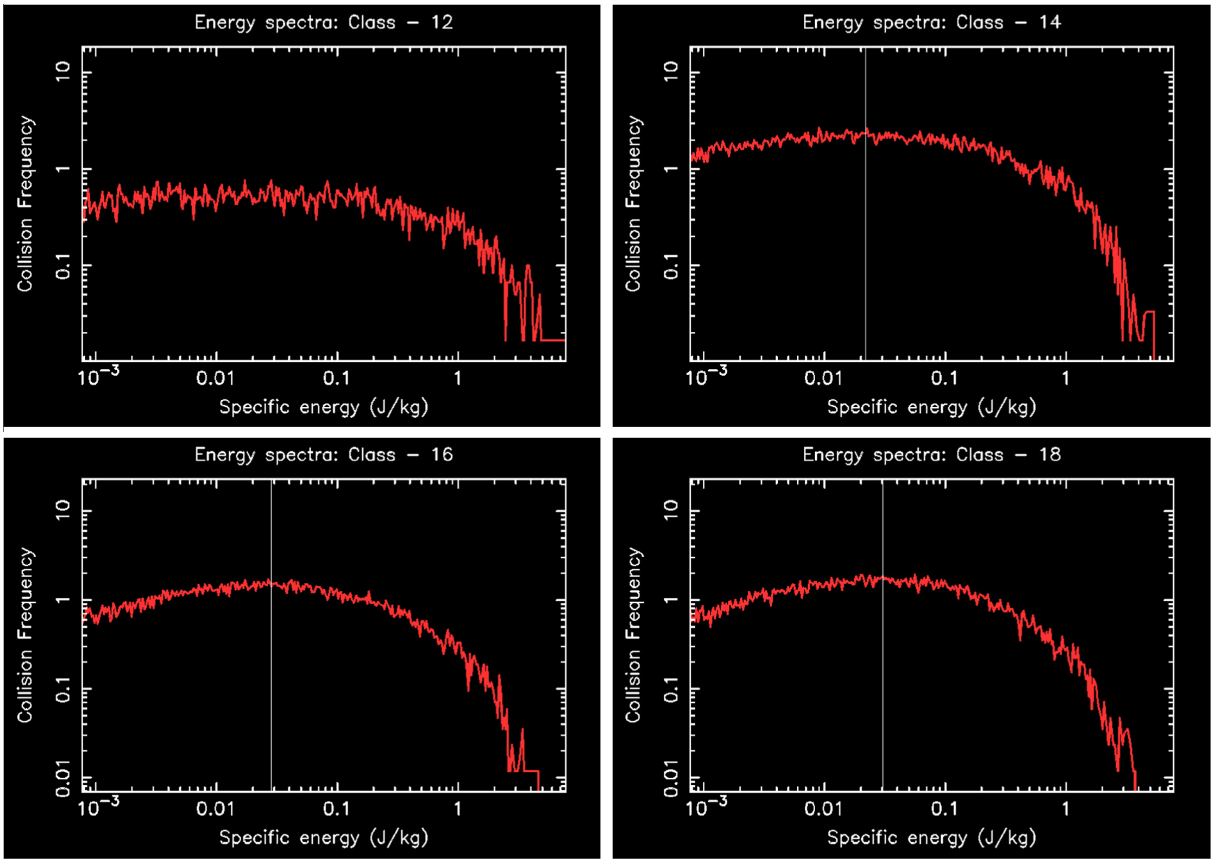

Fig. 15. Upper end of the energy spectra for SAG case 3 for different rock size classes (12 is the smallest and 18 is the highest class shown).

Table 8
Peak collision energy intensity (J/kg) for different rock size classes for AG case 1 and SAG case 3. Peak energies are determined as the largest energy with a frequency of 0.01 Hz or more over the entire 12 min grinding duration.

Class	AG case 1	SAG case 3
12	–	4.5
13	–	4.6
14	3.9	4.6
15	3.6	4.3
16	3.8	3.6
17	3.1	3.4
18	3.2	3.0
19	2.1	1.9

energy spectra. This would cause many more collisions to exceed the elastic threshold and to exceed it by a larger degree leading to much greater incremental damage energy to be accumulated and therefore significant body breakage.

The maximum fall height for a particle in this mill is around 1.2–1.4 m and the majority of particles have masses of 2 kg or less so the maximum kinetic energy of a particle falling in the cataracting stream on impact at the toe is 24–28 J. The lowest coefficient of restitution is for rock–steel collisions with a value of 0.5 being used. This means that around 75% of the initial kinetic energy can be dissipated which is 18–21 J. It is rare for a particle to fall the full distance without some intermediate collisions so the peak energy dissipations observed per particle pair are around 8–9 J which is around half the theoretical limit just calculated. The peak collision energies observed are around double the peaks for the spectra for each size (see Table 8) since we have assumed a 50–50 split of energy dissipation between the colliding bodies. The mill is not able to create collisions with total energy dissipation above about 9 J. So the only way that the breakage observed for SAG case 3 can therefore occur is for the rock particles to absorb a much greater fraction of the energy dissipated in each collision.

As a demonstration of the important effect of the attribution of energy dissipation in collisions to the participating bodies, the AG case 1 and SAG case 3 simulations were run again, this time attributing 90% of the collision energy dissipation to the rock particles. Fig. 16 shows the energy spectra for rock size class 18 (fairly coarse) for (top row) AG case 1 and (bottom row) SAG case 3. The left column shows the spectra for a 50–50 split of energy between balls and rocks while the right column shows the spectra for a 10–90 split in favour of the rock. For the AG case, there is only a small difference which results from the greater energy transfer to the rock particles when they collide with the steel liner. For the SAG case, there is a much stronger effect as the steel ball–rock collisions also now attribute 90% of their collision energy to the rock particles. The shapes of the spectra are very similar but the maximum energies recorded have increased from around 3.7 J to 6.0 J.

Finally, we compare the maximum collision energy intensities observed for the two different levels for the attribution of energy dissipation to the rock for each of the size classes. In AG case 1 (see Table 9) the increase from considering the reduced steel absorption of energy varies from −15% (for class 16) up to 68% (for class 19). The mean change across all the classes is a modest 28%. The change is not larger because all the dominant rock–rock collisions still split their energy dissipation 50–50 between the rock particles. Only the collisions with the steel liner increase the collision energy intensity strongly, but these are a minority of the collisions. The mean peak energy intensity for the AG mode is 4.8 J/kg. If E_0 were instead to be 4.0 J/kg then only a small num-

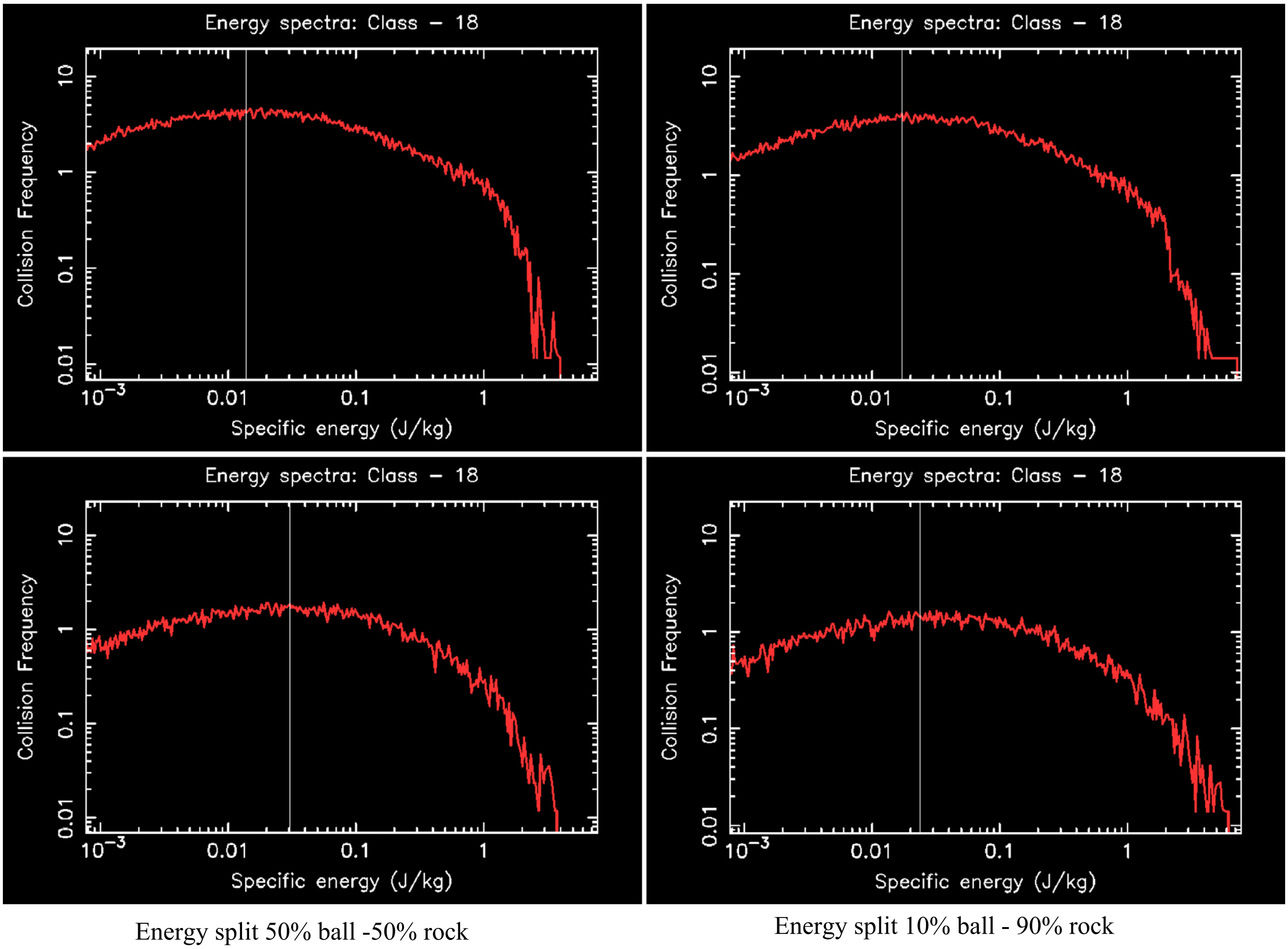

Fig. 16. Energy spectra for rock size class 18 (fairly coarse) for (top row) AG case 1 and (bottom row) SAG case 3. The left column shows the spectra for a 50–50 split of energy between balls and rocks while the right column shows the spectra for a 10–90 split in favour of the rock.

Table 9
Peak collision energy intensity (J/kg) for different rock size classes for the AG case 1 for different levels of energy absorption by the rock. These are calculated over the first minute of grinding operation.

Class	50% absorption	90% absorption	% change
16	5.9	5.0	−15
17	4.0	5.8	45
18	4.0	4.5	13
19	2.2	3.7	68

Table 10
Peak collision energy intensity (J/kg) for different rock size classes for the SAG case 3 for different levels of energy absorption by the rock. These are calculated over the first minute of grinding operation.

Class	50% absorption	90% absorption	% change
16	3.3	6.3	91
17	4.0	6.8	70
18	3.8	6.0	58
19	2.5	4.0	60

ber of collisions would exceed this level and the incremental damage to the rock particles would be minimal. For SAG case 3 (see Table 10) the increase from the reduced steel absorption of energy varies from 58% (for class 18) to 91% (for class 16). The mean change in the maximum collision energy intensity is 70%, which is quite significant. For SAG operation, the collisions with both the steel balls and liner transfer much more energy to the rock particles so the effect is much more significant than for the AG mode. The mean peak energy intensity for the SAG case is 5.8 J/kg. So if E_0 were to be 4.0 J/kg then a reasonable number of collisions would exceed this level and the rate of incremental damage would be much larger than for AG mode or for the SAG mode with a 50–50 attribution of collision energy.

The attribution of a much larger fraction of the collision energy to the rock particles is physically quite reasonable. Rock is an intrinsically much more brittle material and is much more able to crack during collision. The stress waves within the steel are transmitted without damaging the steel and are reflected from the rear of the steel structure back into the rock where it can create fracture damage. The choice of a 90–10 split was made here purely to demonstrate the effect and to show that it can explain the observed experimental behaviour.

In analysis relating to an Ultra Fast Load Cell, Tavares and King (1998) assuming a perfectly elastic non-frictional contact between a rock and impactor, gave the relationship for the fraction of the elastic energy of the contact that was stored in the rock particle as:

$$\varphi = \frac{1}{\frac{k_p}{k_s} + 1}, \quad (14)$$

where k_p is the stiffness of the rock particle and k_s is the stiffness of the steel ball. With typical values of k_s = 200 GPa and k_p = 1–50 GPa, this gives a range for φ of 0.8–0.995. This means that considering only purely elastic effects, the energy split deduced from matching the DEM simulations to the experimental data is the middle of this theoretical range. It should also be noted that for collisions between particles of the same material that Eq. (14) predicts equal elastic energy being stored in each particle which is consistent with historical DEM practice of using a 50–50 split of energy.

However, for a brittle frictional material that fractures during the contact, it is not clear what material properties control this energy split, whether it is Young's modulus (as is the case for the elastic non-frictional material), the brittleness of the material, the restitution coefficient or some combination of other properties. It is important to appreciate that the energy being absorbed by the particles which controls breakage is the energy dissipated within the particles rather than the elastic energy stored (which is at least partially restored to the particle as kinetic energy after unloading of the contact). This energy dissipated is, however, likely to be at least partially dependent on the peak elastic energy or elastic strain. This effect is captured to some degree by the coefficient of restitution which is used in setting the dashpot strength in the DEM contact model (Eqs. (1) and (2)). The underlying mechanisms controlling this energy split and methods for characterising the controlling material properties need further investigation now that its importance in breakage prediction using DEM is understood.

10. Conclusions

A computational method for attributing the energy dissipated in particle collisions to five mechanisms identified as causing size and/or shape change for particles in AG, SAG and coarse feed ball mills has been presented. Chipping and rounding are shape dependent mechanisms that lead to preferential mass loss from corners and edges of particles at much lower energy levels than required for body breakage and which are responsible for rounding or conditioning of particles as well as some size reduction. They are controlled by the energy dissipation at a contact and the location of the contact on the surfaces of the particles (which defines both the contact normal and the moment vector). Single impact body breakage has been found to be a weak contributor to overall size reduction in AG, SAG and coarse feed ball mills. Most body breakage occurs as a result of damage accumulation over many collisions. This damage is proportional to the collision energy above an elastic threshold with energy dissipation below this resulting only in heat and sound. This incremental damage mechanism is inherently less efficient than single body breakage as a much larger fraction of the total input energy is below the elastic threshold. Much of the size reduction is then the result of the accumulation of the damage from many weak collisions experienced by particles within mill. Finally, size reduction from abrasion is well represented by the shear energy absorption of the particles.

The chipping and rounding mechanisms have a significant effect on the distributions of the particle shape attributes. For the three test conditions used in this study of a well characterised dilute pilot AG/SAG mill, we find that the:

- Collisions occur preferentially with corners and frictional contacts also preferentially occur at corners and edges. This concentrates the erosion at these more exposed locations leading to preferential rounding of corners and edges.
- Blocky or angular particles (those with $m > 4.5$) lose their corners and edges more rapidly compared to initially more rounded particles.
- The cumulative distribution of blockiness moves to smaller blockiness values and its gradient increases (meaning that the range of blockiness becomes narrower) as the particles are ground.
- Blockiness changes much more rapidly than the aspect ratios of the particles.
- The rate of aspect ratio change is dependent on the aspect ratio, with particles having aspect ratios of 1.2 or less changing only slowly but with faster reduction rates for higher aspect ratios.
- Changes in the average shape properties for the population of particles in these batch mills is almost linear with time – which means that they essentially proportional to the mass loss rate arising from the chipping and rounding mechanisms. However, the rate of decline of the average blockiness is faster in the early stages but then decreases by around 40% in the latter half of the milling period.

For the dilute batch grinding setup used in this work, the initially dominant rock damage mechanism is rounding with more than a third of the rock damage energy being used to remove the corner material from the particles thereby causing the strong shape change. The chipping mechanism consuming more than 20% of the energy is also quite strong and contributes meaningfully to rock shape and size change. The energy used for incremental breakage is quite small (less than 3% of the rock damage energy for this milling configuration). Wasted energy (which is defined as the energy directed into the particle and which is less than the elastic threshold E_0) and does not contribute to particle damage and can also be estimated using this model. This represents a substantial fraction of the energy consumed at more than 30%. This energy wastage is one of the key reasons for tumbling mill performance being inherently inefficient. For SAG mills with conventional charge size distributions some of this energy will result in the breakage of finer particles (which are absent from this test mill) which could (slightly) reduce the level of energy wastage from the incremental damage mechanism. This possibility is considered further at the end of Section 6.

The rounding or conditioning of the particles (as they are ground) causes more energy to be directed towards their centres. At the end of grinding for 120 min (which is equivalent to the model prediction with the erosion rates scaled by a factor 10) with an AG charge, this leads to a near doubling of the abrasion with sharp corresponding decreases in the amount of rounding. However, the amount of damage energy used for chipping was found to increase strongly (by around 50%). This change in the balance of rounding and chipping (which both depend on particle shape) is a result of the changing balance of collisions being from rock–rock (which has a larger sliding frictional component to their interaction) to rock–liner (which is more predominantly in the normal direction). The strong rounding of the particles at the end of the grinding period means that a larger fraction of the increased energy in the normal direction is directed into the core of the particles causing more collisions to exceed the elastic threshold E_0. This leads to a near doubling in the total amount of incremental damage. For the wasted energy, the increased fraction of normal energy is counter-balanced by the decreased fraction of energy below the elastic threshold, which combined give a very small decrease.

The changes observed over time for a SAG charge are different in pattern and generally smaller in magnitude than for an AG charge. The large increases in abrasion and chipping observed over time for the AG case do not occur for the SAG case and the chipping fraction is essentially constant instead of increasing. The only mechanism that increases more over time for the SAG case was the incremental damage. The grinding media (which does not change size over this grinding period) appears to maintain a more constant grinding environment over time as the rock particles are comminuted. This suggests that full size AG mill performance could be more sensitive to the changing size and shape distributions of the rock within the mill. In contrast, the presence of the media in a SAG mill appears likely to maintain a more constant grinding environment along the mill with the performance perhaps less influenced by the specific nature of the rock charge variation within the mill.

Attrition is the mechanism that is responsible for the shrinkage of large particles bringing them down to the size range at which the mill can start to produce body breakage. The incremental damage is the mechanism then responsible for this body breakage and this is found to be only around 2% for this mill and charge. So the fraction of the energy being used in the two most important mechanisms for significantly reducing the size of rock particles is only 7% of the energy input. Around 43% of the energy is used for removing corners and edges (conditioning). This does contribute to the creation of finer rock fragments but these mechanisms also serve to protect the integrity of the rocks by preventing energy being absorbed by the core of the particles (which is not at all what one would like to happen in a grinding mill).

The inclusion of the size and shape change of particles during grinding within the DEM model also provides information on how these particle changes affect the energy spectra (which are used to characterise the collision environment). There is a general trend of increase in the most commonly generated collision energy per unit mass as the particle size reduces for both AG and SAG charges. This means that as a particle shrinks the proportion of its energy spectra that exceeds the E_0 elastic threshold increases and an increasing fraction of the available collision energy is able to contribute to the incremental damage. So as a particle shrinks in size due to abrasion, chipping and rounding its rate of accumulation of incremental damage increases and its chances of undergoing catastrophic body breakage increases proportionally. This highlights the complex interaction between the different breakage mechanisms that control the evolution life cycle of particles within a tumbling mill.

The charge composition controls the balances of collision types and energy dissipation mechanisms which then control the pattern of size reduction and shape change of the rock charge. The evolving charge constantly changes the balance of these mechanisms with abrasion and chipping increasing but rounding decreasing (mechanisms controlled by the particle shape) and with an increase in the incremental damage energy and a matching reduction in wasted normal impact wasted energy (mechanisms controlled by particle size). The fraction of energy dissipated by the liner is also significant and increases as the rocks reduce in size and collisions occur increasingly with the liner.

The final issue considered by this paper is the effect of the choice of energy attribution between colliding particles. Historically, DEM usage has attributed this equally to each particle. If this was valid then we would observe similar breakage behaviour between the AG and SAG charge cases in the experiments used here which is not the case. The attribution of a much larger fraction of the collision energy to the rock particles is physically quite reasonable. Rock is an intrinsically much more brittle material and is much more able to crack during collision. The stress waves within the steel are transmitted without damaging the steel and are reflected from the rear of the steel structure back into the rock where it can create fracture damage. A semi-arbitrary choice of a 90–10 split between rock and ball was made here and demonstrates that this does indeed explain the observed experimental behaviour. This means that the energy split between particles in DEM simulation of comminution can no longer be ignored. However, it is not fully clear which material properties control this energy split. The underlying mechanisms controlling this energy split and methods for characterising the material properties need significant future investigation.

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